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The Human-Centric Threat Landscape of Ghana's Mobile Financial Ecosystem

Social Engineering, Digital Lending,
and the Regulatory Response

**A Strategic Threat Landscape Report
for the Financial Services Sector**

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Executive Summary

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For the Attention of the Chief Information Security Officer (CISO)

Purpose: This report provides a strategic, intelligence-driven assessment of the human-centric cyber threat landscape targeting Ghana’s mobile financial ecosystem. It examines the convergence of Mobile Money (MoMo) platforms, social engineering attack vectors, the vulnerabilities and predatory practices of the digital lending sector, and the fragmented regulatory framework designed to govern them.

Scope: The assessment focuses on threats originating from and targeting the Ghanaian mobile financial sector. In-scope areas include MoMo platforms (MTN MoMo, Telecel Cash, AT Money), digital lending applications, social engineering tactics (smishing, vishing, Business Email Compromise, WhatsApp impersonation, insider grooming), and the regulatory mandates of the Bank of Ghana (BoG), National Communications Authority (NCA), and Data Protection Commission (DPC). Traditional banking threats, ransomware on core banking systems, SWIFT network attacks, and crypto-asset platforms are excluded.

Methodology: This report is the product of systematic Open Source Intelligence (OSINT) collection and analysis, including review of Ghanaian media archives, Bank of Ghana circulars, Data Protection Commission statements, Interpol African Cyberthreat Assessments, and the Verizon Data Breach Investigations Report (DBIR). Threat techniques are mapped to the MITRE ATT&CK framework. The analysis is further informed by a systematic review of 24 academic and industry sources, including Ghana-specific empirical studies, regional comparators, and global technical frameworks.

Key Findings:

1. **Social engineering is not merely an attack vector; it is *the* primary attack vector.** Analysis indicates that over 90% of mobile financial fraud in Ghana is enabled by some form of human manipulation, not by the exploitation of technical vulnerabilities. The cultural tailoring of these attacks—using local languages, hierarchies, and trust networks—makes them exceptionally effective (3; 19).
2. **The MoMo ecosystem provides a frictionless, low-risk platform for industrial-scale monetization.** The speed of transaction settlement, the irreversibility of trans-

fers, and the vast, loosely monitored agent network create an environment where stolen credentials can be converted into untraceable cash within minutes. Mobile Money accounted for 63.4% of cybercriminal financial flows in Ghana by 2024 (3).

3. **The digital lending sector represents a double threat.** It is both a victim of sophisticated fraud (synthetic identities, loan stacking) and a perpetrator of data weaponization against consumers. The harvesting of contact lists and the practice of debt-shaming documented in comparable African markets (16) constitute a clear violation of data protection principles, yet enforcement remains minimal.
4. **The regulatory landscape is fragmented and reactive.** A liability vacuum exists between the BoG, NCA, and DPC, leaving victims of SIM swap and MoMo fraud without clear, time-bound recourse. Ghana's Cybersecurity Act 2020 (Act 1038) contains no explicit provisions addressing AI-enabled threats such as deepfakes and synthetic identity fraud (18; 3), and the Data Protection Act 2012 (Act 843) suffers from chronic enforcement gaps (24).

Strategic Recommendations (Prioritized):

1. **Establish a National Mobile Financial Fraud Coordination Framework (NM3F).** A joint, real-time task force co-staffed by BoG, NCA, DPC, CSA, MNOs, and financial institutions to enable intelligence sharing, coordinated takedown of phishing infrastructure, and unified public awareness messaging.
2. **Mandate psychological inoculation training across the financial sector.** Replace annual, passive computer-based training with frequent, simulated social engineering campaigns using localized scripts in Ghanaian languages. Extend mandatory certification to the entire MoMo agent network.
3. **Issue a binding Digital Lending Data Ethics Directive.** The Data Protection Commission must explicitly prohibit contact list harvesting and debt-shaming practices, with severe financial penalties and license revocation. The regulatory scope must cover bank-MNO partnership products, not solely standalone fintech lenders (16).
4. **Fix the liability architecture.** BoG and NCA must issue a joint directive establishing clear, time-bound liability for SIM swap and MoMo fraud claims, ensuring victims are made whole within a defined period.
5. **Launch a sustained “Verify Before You Trust” national public education campaign.** A government-funded, multi-platform campaign in all major Ghanaian languages, culturally grounded rather than imported from Western awareness frameworks (19).

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Contents

Executive Summary	i
Table of Contents	vi
List of Figures	vii
List of Tables	viii
Acknowledgements	ix
Disclaimer	x
1 Introduction	1
1.1 Background	1
1.2 Problem Statement	2
1.3 Purpose and Objectives	2
1.4 Scope	3
1.5 Methodology	3
1.5.1 Literature and Open Source Intelligence (OSINT)	3
1.5.2 Threat Classification Framework	4
1.5.3 Regulatory Analysis	4
1.5.4 Limitations	4
1.6 Report Structure	4
2 The Battlefield: Ghana’s Mobile Money Ecosystem	6
2.1 The Scale of the Target	6
2.2 The Ecosystem Architecture	7
2.2.1 The Core Stakeholders	7
2.2.2 Transaction Flow and Vulnerability Points	8
2.3 Inherent Platform Vulnerabilities	9
2.3.1 The SIM as Root of Trust	10
2.3.2 The USSD Protocol: Accessibility Without Security	10
2.3.3 The Agent as an Unmonitored Human Endpoint	11
2.3.4 Transaction Irreversibility and Speed	11
2.4 The Emerging AI Amplification Layer	12
2.5 Chapter Summary	12

3	The Weapon: Social Engineering in the Ghanaian Context	14
3.1	The Primacy of the Human Exploit	14
3.2	A Taxonomy of Ghanaian Social Engineering	14
3.2.1	Smishing: SMS Phishing in the Mobile-First Economy	15
3.2.2	Vishing: Voice Phishing and Authority Exploitation	16
3.2.3	WhatsApp Impersonation and the Trusted Network Exploit	16
3.2.4	Business Email Compromise: The Corporate Authority Exploit	17
3.2.5	Insider Grooming and Collusion: The Authorised Threat	18
3.3	The Psychology of the Ghanaian Victim	19
3.4	The Generative AI Acceleration	20
3.5	Chapter Summary	20
4	The Frontier: The Digital Lending Double Threat	22
4.1	The Rise of Digital Lending in Ghana	22
4.2	Threat One: The Platform as Victim	23
4.2.1	Synthetic Identity Fraud	23
4.2.2	Loan Stacking	23
4.2.3	Credential Stuffing and Account Takeover	24
4.2.4	Referral Program Abuse	24
4.3	Threat Two: The Platform as Predator	24
4.3.1	Excessive Permission Harvesting	24
4.3.2	The Debt-Shaming Playbook	25
4.3.3	Data Exploitation Beyond Shaming	25
4.4	The Convergence: A Unified Criminal Ecosystem	26
4.5	The Regulatory Blind Spot: Lessons from Kenya	26
4.6	Chapter Summary	27
5	The Framework: The Fragmented Regulatory Response	29
5.1	The Regulatory Triangulation Problem	29
5.2	The Liability Vacuum	30
5.3	The Cybersecurity Act, 2020 (Act 1038): Ambition and Implementation	31
5.3.1	Statutory Gaps: The AI Blind Spot	31
5.3.2	Resource Constraints and Enforcement Capacity	31
5.3.3	Digital Dependency and Imported Frameworks	32
5.4	The Data Protection Act, 2012 (Act 843): Enforcement Without Teeth	33
5.5	The Missing Mandates	35
5.5.1	No Mandated Security Awareness Training	36
5.5.2	No Data Permission Rules for Digital Lending Applications	36
5.5.3	No Cross-Sector Incident Response Framework	36
5.6	Chapter Summary	37
6	Strategic Recommendations: Building a Human-Centric Defence	38

6.1	Recommendation 1: Establish a National Mobile Financial Fraud Coordination Framework (NM3F)	38
6.2	Recommendation 2: Mandate Psychological Inoculation Training Across the Financial Sector	39
6.3	Recommendation 3: Issue a Binding Digital Lending Data Ethics Directive	40
6.4	Recommendation 4: Fix the Liability Architecture	41
6.5	Recommendation 5: Launch a National “Verify Before You Trust” Public Education Campaign	43
6.6	Prioritisation and Implementation Roadmap	44
6.7	Chapter Summary	44
7	Conclusion	46

List of Figures

List of Tables

2.1	Structural Vulnerabilities of the Ghanaian MoMo Ecosystem and Their Exploitation Consequences	9
5.1	Ghana’s Mobile Financial Security Regulatory Framework: Mandated vs. Missing Requirements	35

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Disclaimer

This Threat Landscape Report is produced for academic and informational purposes as part of a semester project. The intelligence contained herein is derived from Open Source Intelligence (OSINT) and publicly available documents. While every effort has been made to ensure accuracy, the dynamic nature of the threat landscape means that specific threats, actors, and techniques may evolve rapidly.

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Chapter 1

Introduction

1.1 Background

Ghana’s financial services sector has undergone a profound transformation over the past decade. The proliferation of Mobile Money (MoMo) platforms—led by MTN Mobile Money, Telecel Cash (formerly Vodafone Cash), and AT Money—has redefined the architecture of financial inclusion, extending basic banking services to millions of previously unbanked citizens. As of 2024, active MoMo accounts numbered in the tens of millions, with transaction volumes running into hundreds of billions of Ghana Cedis annually (3). The mobile phone has become, in effect, the primary bank branch for a majority of Ghanaians (21).

This transformation represents a remarkable developmental achievement. However, it also constitutes what Arthur et al. (3) have termed the “Digital Inclusion Paradox”: the same infrastructure that drives financial inclusion simultaneously supplies cybercriminals with financial channels, victim pools, and institutional cover. Mogaji and Nguyen (1) identify a typology of dark-side risks embedded in the very structure of mobile money platforms—risks that emerge not from technical failure but from the relational architecture connecting customers, agents, developers, and regulators.

The human being has emerged as the primary attack surface. Research consistently demonstrates that the overwhelming majority of successful cyber intrusions against financial systems are enabled not by the exploitation of software vulnerabilities but by the manipulation of human psychology (8). As documented in the Verizon Data Breach Investigations Report, 74% of breaches result from human error, privilege misuse, stolen credentials, or social engineering (8). In the Ghanaian context, this figure is likely higher, given the structural vulnerabilities embedded in the MoMo ecosystem: dependence on easily hijacked Subscriber Identity Module (SIM) cards as the root of trust, an Unstructured Supplementary Service Data (USSD) protocol that lacks encryption (5; 6), and a vast, loosely monitored agent network that serves as both a distribution channel for financial services and a cash-out infrastructure for fraud (1).

The threat is further compounded by the rise of generative artificial intelligence. Arthur et al. (3) document a steep increase in AI-enabled fraud within Ghana’s financial

ecosystem: AI elements were present in only 8.3% of cybercrime cases in 2022, but this figure rose to 47.2% by the first half of 2025. AI-enabled cases carry 2.73 times the average financial loss of non-AI cases. Deepfake voice cloning, which exploits Ghana’s culturally embedded voice-based trust norms in agent and peer-to-peer transactions, represents a qualitatively new threat vector for which existing regulatory instruments are unprepared (18).

1.2 Problem Statement

Despite the scale and sophistication of these threats, the response remains fragmented. The regulatory architecture—distributed across the Bank of Ghana (BoG), the National Communications Authority (NCA), the Data Protection Commission (DPC), and the Cyber Security Authority (CSA)—creates accountability gaps that leave victims without clear recourse (3). The digital lending sector, which has grown rapidly as an adjunct to the MoMo ecosystem, operates in a regulatory grey zone where predatory data practices flourish alongside legitimate credit provision (16). Social engineering attacks, culturally tailored to Ghana’s communal trust norms and executed in local languages, continue to succeed at alarming rates because countermeasures remain generic, imported, and psychologically naive (19; 22).

The central problem this report addresses is not the existence of cyber threats—threats are a permanent feature of any digital financial system—but the systematic failure to understand and counter the *human-centric* nature of those threats. The greatest cyber threat to Ghana’s financial sector is not a piece of code. It is the industrialized manipulation of human beings, enabled by vulnerable mobile platforms, exploited by predatory digital lending models, and allowed to fester in a fragmented regulatory landscape.

1.3 Purpose and Objectives

This report provides a strategic, intelligence-driven assessment of the human-centric cyber threat landscape targeting Ghana’s mobile financial ecosystem. Its specific objectives are:

1. To map the MoMo ecosystem as a threat environment, identifying the inherent architectural vulnerabilities that make it susceptible to human-centric attack vectors.
2. To produce a detailed taxonomy of social engineering threats as they manifest in the Ghanaian context, including the psychological, cultural, and linguistic factors that amplify their effectiveness.
3. To examine the digital lending sector as a dual-threat phenomenon: simultaneously a target of sophisticated fraud and a perpetrator of data weaponization against consumers.
4. To analyze the regulatory landscape governing mobile financial security in Ghana, identifying fragmentation points, enforcement gaps, and statutory blind spots.

5. To provide prioritized, evidence-based strategic recommendations for policymakers, financial institutions, and telecommunications operators.

1.4 Scope

This report focuses on the Ghanaian mobile financial ecosystem during the period 2020–2025. The following domains are explicitly within scope:

- Mobile Money platforms operated by MTN Ghana, Telecel Ghana, and AT Ghana.
- Digital lending applications and micro-credit platforms integrated with MoMo infrastructure.
- Social engineering attack vectors, including smishing (SMS phishing), vishing (voice phishing), Business Email Compromise (BEC), WhatsApp impersonation, and insider grooming and collusion.
- The regulatory mandates and operational capacities of the Bank of Ghana, the National Communications Authority, the Data Protection Commission, and the Cyber Security Authority.
- The Cybersecurity Act 2020 (Act 1038), the Data Protection Act 2012 (Act 843), and relevant subsidiary directives including the BoG Cyber and Information Security Directive.

The following domains are explicitly excluded from this assessment: traditional brick-and-mortar bank robberies and physical cash theft; ransomware targeting core banking systems (unless it intersects with social engineering as an initial access vector); SWIFT network-specific attacks; cryptocurrency and virtual asset platforms as a distinct sector; and threats to non-financial Critical Information Infrastructure.

1.5 Methodology

This report employs a multi-method intelligence collection and analysis framework.

1.5.1 Literature and Open Source Intelligence (OSINT)

The primary research method is a systematic review of 24 academic and industry sources, identified through structured searches of Google Scholar, IEEE Xplore, SSRN, and Ghanaian university institutional repositories. The source corpus comprises:

- **Ghana-specific empirical studies:** Eight sources providing original data on Ghana's mobile money ecosystem, regulatory architecture, and cybercrime patterns (3; 12; 14; 17; 18; 21; 24; 19).

- **Regional comparators:** Four sources from Nigeria and Kenya, selected for structural comparability given shared telecommunications–banking MoMo partnership models (1; 6; 16; 19).
- **Global technical and theoretical frameworks:** Twelve sources providing peer-reviewed taxonomies, attack-phase models, cross-cultural trust theory, AI-driven fraud detection frameworks, and critical sovereignty perspectives (4; 5; 7; 8; 9; 10; 11; 13; 15; 20; 22; 23).

1.5.2 Threat Classification Framework

Social engineering threats are classified using the comprehensive taxonomy developed by Zaoui et al. (8), which categorizes attacks by environment (in-person versus remote), approach (physical-based versus psychological trigger-based), and medium (email, SMS, VoIP, website). Attack progression is modeled using the empirically validated eight-phase social engineering attack model of Karadsheh et al. (10). Where applicable, specific attack techniques are mapped to the MITRE ATT&CK framework.

1.5.3 Regulatory Analysis

The regulatory analysis employs doctrinal legal review of the Cybersecurity Act 2020 (Act 1038), the Data Protection Act 2012 (Act 843), and the Bank of Ghana’s Cyber and Information Security Directive, supplemented by empirical assessments of regulatory effectiveness from peer-reviewed sources (17; 14; 18).

1.5.4 Limitations

Several limitations should be noted. Source 3 (3), while the single most comprehensive Ghana-specific source, is a preprint and has not yet undergone formal peer review. Source 2 (2) has been classified as grey literature pending verification of its statistical claims against primary Bank of Ghana and GSMA sources. Source 9 (9) and Source 12 (12) are doctoral dissertations for which only introductory and literature-review chapters were available; their primary empirical findings are not reflected in this report. The Kenya-specific digital lending case study (16) is treated as a structurally transferable comparator, not as direct evidence about Ghanaian lending practices.

1.6 Report Structure

The remainder of this report is organized as follows:

- **Chapter 2** maps the Ghanaian Mobile Money ecosystem as a threat environment, examining its scale, architecture, and inherent vulnerabilities.

- **Chapter 3** provides a detailed taxonomy of social engineering threats in the Ghanaian context, including analysis of the psychological and cultural factors that amplify their effectiveness.
- **Chapter 4** examines the digital lending sector as a dual-threat phenomenon—both a target of sophisticated fraud and a perpetrator of data weaponization against consumers.
- **Chapter 5** analyzes the fragmented regulatory landscape, identifying accountability gaps between the BoG, NCA, DPC, and CSA, and statutory blind spots in Act 1038 and Act 843.
- **Chapter 6** presents prioritized, evidence-based strategic recommendations for policy-makers, financial institutions, and telecommunications operators.
- **Chapter 7** concludes the report with a synthesis of findings and a forward-looking assessment of the evolving threat landscape.

Chapter 2

The Battlefield: Ghana’s Mobile Money Ecosystem

2.1 The Scale of the Target

To understand the threat landscape, one must first understand the terrain. Ghana’s Mobile Money (MoMo) ecosystem is not merely a payment channel; it is, for a majority of Ghanaians, the primary interface with the formal financial system. As of 2024, active Mobile Money accounts in Ghana numbered in the tens of millions, with annual transaction volumes exceeding hundreds of billions of Ghana Cedis (3). The platform has achieved what decades of traditional banking expansion could not: extending financial services to rural populations, informal sector workers, and low-income households at unprecedented scale and speed (21).

However, this success carries a structural corollary that defines the threat landscape. Arthur et al. (3) have termed this the “Digital Inclusion Paradox”: the very infrastructure that drives financial inclusion simultaneously supplies cybercriminals with financial channels, victim pools, and institutional cover. The statistics are stark. Mobile Money accounted for 63.4% of cybercriminal financial flows in Ghana by 2024, up from 41.2% in 2021. Fraud incidents grew by 214% between 2020 and 2024, outpacing even the 163% growth in legitimate transaction volume over the same period (3). The platform that has become the engine of financial inclusion has also become the vehicle of choice for cyber-enabled financial crime.

The concentration of value within the ecosystem makes it a high-value target. Bank of Ghana data analysed by Opuni-Frimpong et al. (17) reveals that reported cyber fraud cases in the financial sector rose from GHC 3,251,780 in 2021 to GHC 68,699,830 in 2022—an increase of approximately 21 times in a single year. While not all of this fraud transits through MoMo channels, the mobile platform’s dominance in retail payments means it absorbs a disproportionate share of criminal attention. The financial sector attracts 18% of cybercriminals’ activity (17), and within that sector, the mobile money channel—by virtue of its reach, speed, and structural vulnerabilities—is the path of least resistance.

2.2 The Ecosystem Architecture

The Ghanaian Mobile Money ecosystem is not a monolithic system operated by a single entity. It is a complex, multi-stakeholder architecture whose very structure creates security challenges. Mogaji and Nguyen (1) provide a useful typology of four stakeholder relationships, each with its own dark-side risks: customer–developer, customer–agent, developer–agent, and developer–regulator. Mapping these relationships onto the Ghanaian context reveals the architectural seams that threat actors exploit.

2.2.1 The Core Stakeholders

Mobile Network Operators (MNOs). MTN Ghana dominates the market with approximately 58% of mobile subscriptions, followed by Telecel Ghana (formerly Vodafone Ghana) and AT Ghana (formerly AirtelTigo). These MNOs are not merely telecommunications providers; they are, in effect, financial institutions operating under a regulatory framework that was not originally designed for financial services. Their Mobile Money platforms—MTN MoMo, Telecel Cash, and AT Money—hold customer funds in pooled trust accounts at partner banks, but the operational control of the platform, including transaction authorisation and customer authentication, rests with the MNO.

Partner Banks and Settlement Institutions. Each MNO partners with multiple commercial banks that hold the pooled trust accounts and provide liquidity management. The Ghana Interbank Payment and Settlement Systems (GhIPSS) provides the interoperability layer that enables real-time transactions between different MoMo platforms and between MoMo wallets and traditional bank accounts. GhIPSS is, from a threat perspective, a single point of systemic failure: a successful attack on its infrastructure would not compromise one institution but would freeze the entire national mobile payment system.

The Agent Network. Perhaps the most distinctive feature of the Ghanaian MoMo ecosystem—and its most significant human vulnerability—is the agent network. Over 300,000 registered agents operate as the physical interface between the digital platform and the cash economy. Agents perform customer registration, Know Your Customer (KYC) verification, cash-in and cash-out transactions, and PIN resets. They are typically small business owners operating on thin margins, with limited formal training in cybersecurity or fraud detection (1). From a threat actor’s perspective, each agent represents a human endpoint with the authority to authorise transactions, access customer data, and—crucially—bypass digital security controls through manual overrides.

The Customer. The end user of the MoMo ecosystem is, in the majority of cases, a financially vulnerable individual with limited digital literacy. Anane-Donkor and Dei (21) found in a survey of 800 respondents across four institution types that rural and community bank (RCB) clients consistently reported the weakest outcomes across nearly every dimension of digital financial service experience—including customer support and dispute resolution (mean score 3.70 out of 5), responsiveness to downtime and errors (3.60), and credit access improvement (3.70). These are the populations most dependent on MoMo

for daily financial transactions and, simultaneously, the least equipped to recognise or resist social engineering attacks.

2.2.2 Transaction Flow and Vulnerability Points

A typical MoMo transaction follows a deceptively simple path: the customer initiates a transfer via USSD (e.g., *737# for MTN MoMo) or a smartphone application, specifying the recipient's phone number and the amount. The MNO's platform authenticates the customer—typically via a Personal Identification Number (PIN)—verifies the availability of funds, and executes the transfer. If the recipient is on the same MNO network, the funds are credited instantly. If the recipient is on a different MoMo platform or a bank account, the transaction routes through GhIPSS.

At each stage of this flow, vulnerability points exist. The USSD channel, which remains the dominant access method for feature phone users and rural populations, transmits data in plain text with no encryption (5). The PIN, entered on the device, can be observed through shoulder surfing—a risk that Anunobi et al. (5) found could be reduced to near zero through partial-PIN challenge systems, but which remains unaddressed in the standard Ghanaian MoMo implementation. The phone number, which serves as the account identifier, is also the primary target for SIM swap attacks (4).

2.3 Inherent Platform Vulnerabilities

Table 2.1: Structural Vulnerabilities of the Ghanaian MoMo Ecosystem and Their Exploitation Consequences

Vulnerability	Description	Exploitation Consequence
SIM as Root of Trust	The phone number serves as the account identifier, and possession of the SIM card is treated as proof of identity for authentication and transaction authorisation (7).	SIM swap fraud enables complete account takeover. Threat actors intercept OTPs, reset MoMo PINs, and drain wallets and linked bank accounts within minutes. SIM swap accounts for 19.7% of all MoMo fraud in Ghana (3).
USSD Protocol Limitations	USSD transmits data in plain text with no encryption. PINs are entered unmasked on screen. The protocol’s 120-second session timeout constrains authentication complexity (5; 6).	Shoulder surfing enables trivial PIN capture. Man-in-the-middle attacks on USSD sessions are feasible with relatively low technical sophistication. Credential interception enables fraudulent transactions.
Agent as Unmonitored Human Endpoint	Over 300,000 agents operate as the physical interface between digital currency and cash. Agents perform KYC verification, PIN resets, and cash-out transactions with limited cybersecurity training or monitoring (1).	Agent collusion enables industrial-scale cash-out of fraudulent transactions. Agents bypass KYC checks for bribes. Customer data is divulged to third parties. Only 1,200 agents (0.4%) were blacklisted in 2024 (2).
Transaction Irreversibility and Speed	MoMo transactions settle instantly and are effectively irreversible. No recovery window exists between fraud detection and fund extraction (3).	“Cappuccino” cash-out networks drain compromised accounts across dozens of agent points simultaneously within minutes. Automated monitoring systems cannot distinguish fraud from legitimate high-volume activity in real time.
Generative AI Amplification	AI tools enable deepfake voice cloning, synthetic identity generation, and automated production of culturally tailored phishing content at scale (3; 20).	AI-enabled attacks carry 2.73× the financial loss of conventional attacks. AI elements were present in 47.2% of cybercrime cases by H1 2025. KYC verification systems are defeated by AI-generated Ghana Card images and biometric data.

The vulnerabilities of the MoMo ecosystem are not bugs in an otherwise sound system. They are emergent properties of design choices that prioritised speed, accessibility, and

financial inclusion over security. Four structural vulnerabilities stand out as the primary enablers of the human-centric threats analysed in subsequent chapters.

2.3.1 The SIM as Root of Trust

The foundational security assumption of the MoMo ecosystem is that possession of a SIM card constitutes proof of identity. The phone number is the account number. The SIM card, inserted into a handset, is the authentication token. A One-Time Password (OTP) sent via SMS to that number is treated as a second factor. This architecture, as Mutabazi et al. (7) argue, “implicitly equates SIM possession with identity”—it trusts the SIM card, not the human being.

The consequence is that SIM swap fraud—in which a threat actor fraudulently transfers a victim’s phone number to a SIM card in their possession—is not an exotic attack requiring advanced technical skills. It is a social engineering attack against a telecommunications provider’s customer service process (4). Kim et al. (4) document the standard three-step process: personal data theft (gathering enough information about the victim to pass authentication checks), fraudulent SIM re-issuance using that data, and exploitation of the hijacked line to access mobile banking, wallets, and other accounts via intercepted OTPs. Once the SIM is swapped, the threat actor owns the victim’s entire mobile financial identity.

The scale of this vulnerability in Ghana is substantial. Arthur et al. (3) identify SIM swap as the second most prevalent MoMo fraud vector, accounting for 19.7% of all mobile money fraud incidents. The same source notes that 58–72% of mobile money fraud in the studied African countries is attributable to social engineering, with SIM swap and agent-assisted fraud as the two dominant vectors (2). The SIM card, intended as a tool of inclusion, becomes the single point of catastrophic failure.

2.3.2 The USSD Protocol: Accessibility Without Security

The Unstructured Supplementary Service Data (USSD) protocol is the technological backbone of Ghana’s mobile money revolution. It works on every handset, from the most basic feature phone to the latest smartphone. It requires no internet connection. It is fast, familiar, and functionally universal.

It is also, from a security perspective, fundamentally inadequate for financial transactions. USSD transmits data in plain text with no encryption (5). PINs are entered unmasked on the screen, making shoulder surfing a trivial attack vector (6). Lamoyero and Fajana (6), in their systematic audit of 19 commercial banks’ USSD authentication protocols in Nigeria—a directly comparable market given shared MNO infrastructure—found that approximately 70% of banks required no authentication at all for balance enquiries and airtime purchases. Only 4 of 19 banks met the NIST AAL2 (two-factor authentication) standard for PIN resets. PINs were stored and transmitted unmasked, and several banks used easily obtainable personal data such as date of birth as a “second

factor”—which is not independent authentication at all.

These findings are not Nigeria-specific curiosities. They reflect the structural limitations of the USSD protocol itself, limitations that apply equally to Ghana’s MTN, Telecel, and AT Money platforms. The protocol’s 120-second session timeout constrains the complexity of authentication challenges that can be implemented (5). The text-only interface cannot support biometric verification or graphical password schemes. The protocol was designed for simple menu-based services, not for securing financial transactions at national scale.

2.3.3 The Agent as an Unmonitored Human Endpoint

The MoMo agent occupies a unique and precarious position in the ecosystem’s security architecture. Agents are simultaneously the most critical human interface for customer trust and the most exploited vulnerability for fraud execution (1). Mogaji and Nguyen (1) found that the customer–agent relationship was the strongest relational bond in the ecosystem, but also the most rife with fraud risk. Agents in the Nigerian context were reported to charge above-stipulated fees, divulge customers’ transaction details and addresses to third parties—two customers reported being burgled after visiting an agent—and possibly collude with banks to keep ATMs empty and force Point-of-Sale usage.

In the Ghanaian context, the agent’s power is amplified by the functions they perform. Agents conduct KYC verification during customer registration, creating a gatekeeper role that can be exploited for synthetic identity creation. They process cash-out transactions, which is the final and most critical stage of the fraud lifecycle—the point at which stolen digital funds become untraceable physical cash. They handle PIN resets, which is functionally equivalent to password resets in a traditional banking context but without the multi-layered verification a bank branch would require.

The Bank of Ghana has recognised this vulnerability, reportedly blacklisting 1,200 “bogus agents” in 2024 (2). However, with over 300,000 registered agents, the blacklisting of 1,200 represents just 0.4% of the total agent population. The structural problem remains: the agent network was designed for distribution and accessibility, not for security. Each agent is a human firewall, and the majority have received no formal cybersecurity training.

2.3.4 Transaction Irreversibility and Speed

A MoMo transaction settles in seconds. Unlike a bank wire transfer, which may take hours or days and can potentially be reversed if fraud is detected quickly, a MoMo transfer is effectively instantaneous and irreversible. This design choice serves legitimate users well—it enables the real-time payments that underpin Ghana’s dynamic informal economy—but it also serves fraudsters. The window between a victim realising they have been defrauded and the funds being irretrievably cashed out at multiple agent points is

measured in minutes, not hours (3).

The cash-out infrastructure is optimised for this speed. Arthur et al. (3) document the “cappuccino” cash-out model, in which networks of complicit agents process high-volume, high-speed withdrawals from compromised accounts across dozens of locations simultaneously. The transaction volume makes the event look like a busy business day to automated monitoring systems. By the time anomaly detection algorithms flag the pattern, the funds are gone.

2.4 The Emerging AI Amplification Layer

A threat landscape report written in 2024 would have identified the four vulnerabilities above as the primary structural enablers of mobile money fraud. Writing in 2026, it is necessary to add a fifth dimension that is qualitatively transforming the threat environment: the role of generative artificial intelligence as an amplifier of existing vulnerabilities.

Arthur et al. (3) provide the most comprehensive Ghana-specific data on this phenomenon. AI-enabled elements were present in only 8.3% of cybercrime cases in Ghana in 2022. By the first half of 2025, this figure had risen to 47.2%. Critically, AI-enabled cases carry 2.73 times the average financial loss of non-AI cases. The technology is not merely increasing the volume of attacks; it is increasing their lethality.

The specific AI applications most relevant to the MoMo ecosystem include deepfake voice cloning—which exploits Ghana’s culturally embedded voice-based trust norms in agent and peer-to-peer transactions—and synthetic identity generation capable of defeating KYC and Ghana Card verification systems (3; 20). Sambo et al. (15) confirm that deep learning-based fraud detection systems achieve F1 scores of approximately 0.90, with Graph Neural Network and Federated Learning hybrids reaching approximately 0.94—but these advanced systems carry computational costs and latency requirements that may exceed the capabilities of smaller Ghanaian financial institutions, creating a defensive asymmetry that favours attackers.

2.5 Chapter Summary

The Ghanaian Mobile Money ecosystem is a remarkable achievement of financial inclusion, but it is also a threat environment whose structural characteristics make human-centric attacks both easy and profitable. The SIM card serves as the root of trust despite being trivially hijackable through social engineering. The USSD protocol provides universal access at the cost of fundamental security. The agent network, 300,000 strong, constitutes an unmonitored human endpoint with transaction authority and access to customer data. Transaction irreversibility and settlement speed eliminate the recovery window that traditional banking fraud departments rely upon. And generative AI is now amplifying each of these vulnerabilities, making attacks more convincing, more scalable, and more financially damaging.

This is the battlefield. The weapon deployed upon it—social engineering—is the subject of the next chapter.

Chapter 3

The Weapon: Social Engineering in the Ghanaian Context

3.1 The Primacy of the Human Exploit

If the Mobile Money ecosystem is the battlefield, social engineering is the weapon of choice. It is a weapon that requires no code, no exploit kit, and no technical sophistication. It requires only an understanding of human psychology—and, in the Ghanaian context, an understanding of local languages, cultural norms, and communal trust structures that make the population uniquely susceptible to specific forms of manipulation.

The global data establishes the baseline. The Verizon Data Breach Investigations Report documents that 74% of breaches result from human error, privilege misuse, stolen credentials, or social engineering, with 83% involving external actors driven by financial motive in 95% of cases (8). Chhabra Roy and Prabhakaran (11) note that 58% of attacks on financial institutions rely on employees to gain access—but over 90% of those employees are unwitting social engineering victims, not malicious insiders. The human being, not the firewall, is the perimeter that attackers target.

In the Ghanaian mobile financial ecosystem, the economics of social engineering are particularly favourable to attackers. A sophisticated zero-day exploit against a banking application might cost tens of thousands of dollars on the dark market and be patched within days. A well-crafted smishing message in Twi, sent to 10,000 MTN MoMo users, costs virtually nothing and can be redeployed indefinitely with minor variations. The return on investment is asymmetric. Arthur et al. (3) confirm that agent impersonation alone accounts for 28.4% of all mobile money fraud in Ghana, making it the single largest fraud category, ahead of SIM swap (19.7%), fake customer care fraud (16.3%), and reversal fraud and Business Email Compromise (22.1%).

3.2 A Taxonomy of Ghanaian Social Engineering

Zaoui et al. (8) provide a comprehensive, peer-reviewed taxonomy that classifies social engineering attacks by environment (in-person versus remote), approach (physical-based

versus psychological trigger-based), and medium (email, SMS, VoIP, website, physical access). This taxonomy, combined with the empirically validated eight-phase attack model of Karadsheh et al. (10)—identification of potential target, information aggregation, analysis and interpretation, target recognition, decision approach, armament, influence, and execution—provides the theoretical framework for the Ghana-specific analysis that follows.

What emerges when Ghanaian threat data is mapped onto these frameworks is not a generic global pattern but a culturally specific threat landscape. Adewopo et al. (19) argue that social engineering in West Africa exploits “cultural values, local languages, and communal trust systems” in ways that imported Western prevention frameworks fail to address. The following subsections dissect each major attack vector as it manifests in Ghana.

3.2.1 Smishing: SMS Phishing in the Mobile-First Economy

Smishing—phishing conducted via SMS—is the most volumetrically significant social engineering vector in Ghana’s mobile financial ecosystem. Its effectiveness derives from the convergence of three factors: the universality of SMS as a communication channel (every MoMo user, by definition, has an active phone number), the trusted position of MNOs as the senders of legitimate service messages (balance alerts, transaction confirmations, promotional offers), and the limited screen real estate of mobile devices, which makes it harder for users to inspect URLs or detect spoofed sender identifiers.

Adongo Mendel (2) documents the standard smishing templates circulating in Ghana and the broader West African region. Attackers spoof MTN, Telecel, and AT Money branding, crafting messages that create urgency through threat of account suspension: “Your MoMo account has been flagged for security review. Click here to reactivate: [malicious link].” Seasonal variants exploit predictable financial pressure points—school fee payment periods, Christmas and Easter spending, and tax filing deadlines. The messages are increasingly composed in Twi, Ewe, Ga, and Hausa, as well as Ghanaian Pidgin English, reflecting a shift toward hyper-localised targeting that generic, English-language spam filters are ill-equipped to detect.

The linguistic sophistication of these campaigns should not be underestimated. A 2024 smishing campaign documented in Ghanaian media used grammatically perfect Twi with appropriate honorifics and cultural references, making it indistinguishable from legitimate MTN customer communications for users whose primary language is Twi and whose exposure to formal cybersecurity awareness materials—overwhelmingly produced in English—is minimal.

The technical infrastructure supporting smishing campaigns is similarly low-cost and difficult to disrupt. Attackers use bulk SMS gateways, often routed through multiple jurisdictions, and phishing pages hosted on compromised legitimate domains or short-lived disposable domains. The USSD protocol’s lack of encryption (5) means that even if a user becomes suspicious after clicking a link, the initial credential capture has already

occurred.

3.2.2 Vishing: Voice Phishing and Authority Exploitation

If smishing is the volume play, vishing—voice phishing conducted via phone calls—is the precision strike. Vishing exploits the most fundamental human instinct in the Ghanaian social context: deference to authority.

The standard vishing script follows a predictable pattern, documented across multiple Ghanaian fraud reports and confirmed by Arthur et al. (3). A victim receives a call from an individual claiming to be an MTN MoMo customer service representative, a Bank of Ghana official, or—in more sophisticated variants—a police officer investigating a fraud case. The caller informs the victim that their account has been compromised, that a suspicious transaction has been detected, or that their account requires a “security upgrade.” The victim is instructed to provide their MoMo PIN, confirm a One-Time Password, or—in a particularly insidious variant—dial a specific USSD code that initiates a transfer while the victim believes they are verifying their identity.

The caller’s fluency in the victim’s local language is a critical success factor. Adewopo et al. (19) document the “culture-centered” nature of West African social engineering, noting that attackers leverage local cultural values—respect for elders and authority figures, communal obligation, and religious faith—to lower the victim’s critical evaluation of the request. A vishing call in Twi from someone who sounds like a respected authority figure, who knows the victim’s name and phone number, and who speaks with the calm assurance of an official, is a far more potent weapon than a generic English-language phishing email.

The “reversal fraud” variant deserves particular attention. In this scenario, the fraudster sends a small amount of mobile money to the victim—often GHS 10 or GHS 20—and then calls, pleading poverty, claiming the transfer was a mistake, and begging the victim to return the money. If the victim complies by sending money back, they have authorised a transaction from their own wallet. The fraudster then uses a different channel to file a formal complaint and obtain a reversal of the *original* transaction, effectively doubling their money. This attack exploits two psychological principles simultaneously: reciprocity (the victim feels obligated to return the “mistaken” transfer) and empathy (the fraudster’s pleading triggers a compassionate response). Both principles are documented in Cialdini’s six principles of persuasion, which Karadsheh et al. (10) empirically validate as the psychological foundation of social engineering success.

3.2.3 WhatsApp Impersonation and the Trusted Network Exploit

WhatsApp is not merely a messaging application in Ghana; it is a primary social infrastructure. Family groups, church groups, professional networks, and community associations all operate through WhatsApp. The platform’s end-to-end encryption, while

a privacy feature for legitimate users, makes it an ideal operational environment for social engineers—communications cannot be intercepted or monitored by external security systems.

Three WhatsApp-specific attack patterns dominate the Ghanaian threat landscape.

Family Impersonation. The “Mum/Dad, I lost my phone, this is my new number, please send me money for an emergency” script is globally common, but its effectiveness in Ghana is amplified by the strength of familial obligation and the prevalence of internal and external migration. A parent in Accra whose child is at university in Kumasi, or a sibling working in Takoradi whose brother has migrated to Europe, is primed to respond urgently to a distress message from a “family member.” The fraudster needs only a phone number and a plausible relationship narrative, both of which can be harvested from social media or previous data breaches.

Compromised Account Takeover. When a WhatsApp account is compromised—typically through a SIM swap that captures the WhatsApp verification code—the attacker gains access to the victim’s entire contact list and chat history. They can then broadcast a distress message to every contact, exploiting the trust those contacts have in the compromised account. Because the message comes from a known number with an established chat history, standard warning signs of fraud are absent. Spangler (13) notes that BEC attacks succeed precisely because the compromised account is genuinely legitimate—standard technical security controls are already bypassed by the time the message reaches the recipient.

Romance Fraud and “Pig Butchering.” Nartey et al. (14) document the Hajia4Reall case as a landmark example: Mona Faiz Montrage, a Ghanaian social media influencer, was prosecuted by the United States Department of Justice for receiving over US\$2 million from elderly American victims through romance scams. The Hajia4Reall case illustrates the transnational dimension of Ghanaian social engineering, but the same techniques are deployed domestically. Long-con romance scams, in which a fraudster builds a trusting relationship over weeks or months before introducing a fraudulent investment opportunity (the “pig butchering” model), are increasingly documented in Ghanaian urban centres. The victim is emotionally invested by the time the financial ask is made, and the psychological barrier to refusal is high.

3.2.4 Business Email Compromise: The Corporate Authority Exploit

Business Email Compromise (BEC) is the most financially damaging single category of social engineering globally. The United States Federal Bureau of Investigation’s Internet Crime Complaint Center received 19,369 BEC complaints in 2020, with total losses exceeding US\$1.8 billion across 177 countries (13). The average per-incident loss was US\$70,960, with some large-company losses exceeding US\$2.5 million.

In the Ghanaian context, BEC exploits a specific cultural vulnerability: the hierarchical, deference-based structure of corporate authority. Spangler (13) defines the standard

BEC pipeline: social-media reconnaissance of the target and close associates, crafting a convincing custom pretext, account hijacking via phishing, and exploiting the compromised account for further fraud. When the impersonated authority figure is the Managing Director or a senior partner, and the recipient is a finance officer in a corporate culture where questioning a superior's instruction is culturally discouraged, the attack's success probability is high.

Carter (23) documented that as consumer bank fraud becomes harder due to improved defences, fraudsters increasingly target business customers via BEC—more than US\$5 billion had been stolen through BEC attacks globally by the time of that report's writing. The Ghanaian financial sector, with its mix of large multinational banks and smaller indigenous institutions, presents a varied attack surface. Larger banks have invested in email security gateways and sender verification protocols, but smaller institutions—rural banks, microfinance institutions, and savings and loans companies—often lack these defences, making them attractive targets.

The “supplier account change” variant is particularly effective. A legitimate supplier's email is compromised and monitored. When an invoice is due, the attacker sends new payment instructions with “updated bank details.” The payment is made to the attacker's account. Because the initial relationship is genuine, the fraudulent instruction does not trigger the suspicion that a cold-contact scam would.

3.2.5 Insider Grooming and Collusion: The Authorised Threat

The most difficult social engineering threat to detect and counter is the one that originates from within the trusted perimeter. Chhabra Roy and Prabhakaran (11) identify 14 categories of insider-led cyber fraud across banking systems and find, through machine learning analysis of nine years of Reserve Bank of India fraud-loss data, that spoofing, identity theft, and hacking are the three highest-severity, most consistently damaging fraud types.

In the Ghanaian MoMo context, the insider threat is concentrated at the agent level. Mogaji and Nguyen (1) document agent collusion as a structural feature of the ecosystem: agents charging above-stipulated fees, divulging customers' transaction details and addresses to third parties, and possible agent–bank collusion to manipulate service availability. The economic pressure on agents—who operate on thin margins and handle large volumes of cash—creates a vulnerability that external fraudsters systematically exploit.

The grooming process follows a predictable pattern. A fraudster identifies an agent, often through existing criminal networks. Small bribes are offered for minor violations—overlooking a KYC deficiency, processing a transaction without full verification. Once the agent is compromised, the requests escalate. The agent becomes a cash-out node in a broader fraud network, processing high-volume withdrawals from compromised accounts in exchange for a percentage of the proceeds. Arthur et al. (3) describe the “cappuccino” cash-out model, in which networks of complicit agents process simultaneous withdrawals across dozens of locations, making detection through transaction monitoring

extremely difficult.

The Bank of Ghana’s reported blacklisting of 1,200 agents in 2024 (2) represents a regulatory acknowledgement of this threat, but the number—representing 0.4% of the total agent population—suggests that enforcement reaches only the most egregious cases. The structural incentives that make agent collusion profitable remain largely unaddressed.

3.3 The Psychology of the Ghanaian Victim

Why does social engineering work so effectively in Ghana? The technical vulnerabilities of the MoMo ecosystem provide the opportunity. The psychology of the victim provides the mechanism.

Alhasan (9) provides a theoretical foundation that is directly applicable to the Ghanaian context. Trust, Alhasan argues, drawing on cross-cultural management theory, is not a universal construct. “Cognitive trust” cultures—typified by the United States, Germany, and Australia—build trust on demonstrated competence and track record. “Affective” or “relational trust” cultures—in which Nigeria is explicitly named, and into which Ghana’s communal social structures fit—build trust through shared experience and emotional and relational bonds.

This distinction is not academic. It has operational implications for why social engineering works differently in Ghana than in Western contexts. A fraudster impersonating a community chief soliciting mobile money donations—a pretext documented by Adongo Mendel (2) as specifically Ghanaian—exploits relational trust. The victim is not evaluating the chief’s “track record” of financial probity; they are responding to a communal and hierarchical obligation. A smishing message that invokes religious obligation—“Send your seed money and God will multiply it”—exploits a trust system rooted in faith rather than verification.

Adewopo et al. (19) argue that this cultural dimension requires a fundamentally different approach to countermeasures. Western awareness campaigns emphasise individual scepticism and verification—“trust no one, verify everything.” In a communal trust culture, this message is not merely ineffective; it is culturally alien. Adewopo et al. propose a “culture-centered” prevention approach that leverages local cultural values and ethics rather than importing Western awareness frameworks. This insight—that the defence must be as culturally specific as the attack—is a recurring theme in the Ghana-specific literature and will inform the recommendations in Chapter 6.

Additional psychological factors compound the vulnerability. Arthur et al. (3) found that rural users experience mobile money fraud at higher rates than urban users (34.2% versus 22.7%) and are far more likely to reduce or stop MoMo usage after exposure (47.3% versus 18.9%). Women report significantly lower trust in MoMo (mean 3.91 versus 4.67 on a 7-point scale) even controlling for direct victimisation, driven by being specifically targeted for urgent-request and romance-fraud social engineering. These findings suggest that the psychological burden of social engineering is not evenly distributed: the most financially vulnerable populations are also the most psychologically impacted.

The digital inclusion paradox thus acquires a psychological dimension. The same populations that MoMo was designed to include—rural, lower-income, less formally educated—are the populations most susceptible to social engineering and most likely to disengage from digital finance after being victimised. The threat does not merely steal money; it erodes the trust upon which financial inclusion depends.

3.4 The Generative AI Acceleration

Any analysis of social engineering written in 2026 must account for the role of generative artificial intelligence, not as a future risk but as a present reality. Arthur et al. (3) document that AI-enabled elements were present in 47.2% of cybercrime cases in Ghana by the first half of 2025, and that AI-enabled cases carry 2.73 times the average financial loss of non-AI cases.

The specific AI applications most relevant to social engineering include:

- **Deepfake voice cloning:** Fraudsters can now generate convincing audio of a known individual's voice from a few seconds of sample audio—material readily available from WhatsApp voice notes, social media videos, or phone call recordings. In a culture where voice-based trust is central to agent and peer-to-peer transactions, the ability to clone a trusted voice represents a paradigm shift in the threat landscape.
- **AI-generated phishing content:** Large language models can produce grammatically perfect, culturally nuanced phishing messages in Twi, Ewe, Ga, Hausa, and Ghanaian Pidgin at unlimited scale. The linguistic errors that once served as warning signs of fraud are eliminated.
- **Synthetic identity generation:** AI systems can generate realistic Ghana Card images, biometric data, and supporting documentation capable of defeating KYC verification systems (20).

Mensah et al. (18) note that Ghana's Cybersecurity Act 2020 (Act 1038) contains no explicit provisions addressing AI-enabled threats—a regulatory gap that will be examined in detail in Chapter 5.

3.5 Chapter Summary

Social engineering in Ghana's mobile financial ecosystem is not a generic, globally uniform threat. It is a culturally tailored, linguistically precise, and psychologically sophisticated attack methodology that exploits specific vulnerabilities in Ghanaian social structures. Smishing campaigns in local languages, vishing calls that weaponise deference to authority, WhatsApp impersonation that exploits communal trust networks, BEC attacks that leverage corporate hierarchy, and insider collusion that monetises economic pressure—each of these vectors succeeds not because Ghanaian users are uniquely careless,

but because the attacks are designed to exploit specifically Ghanaian psychological and cultural terrain.

The global frameworks provided by Zaoui et al. (8) and Karadsheh et al. (10) offer the vocabulary to classify these attacks. The Ghana-specific data provided by Arthur et al. (3), Nartey et al. (14), and Adewopo et al. (19) ground that vocabulary in local reality. The cross-cultural trust theory of Alhasan (9) explains why these attacks work at a psychological level. And the emerging AI data demonstrates that the threat is not static—it is accelerating in both volume and sophistication.

The weapon has been dissected. The next chapter examines one of its most insidious applications: the digital lending sector, where social engineering and data weaponization converge to create a double threat.

Chapter 4

The Frontier: The Digital Lending Double Threat

4.1 The Rise of Digital Lending in Ghana

The same mobile money infrastructure that has revolutionised payments in Ghana has given birth to a parallel revolution in consumer credit. Digital lending platforms—mobile applications that offer instant, unsecured, short-term loans disbursed directly to MoMo wallets—have proliferated rapidly since 2020. Platforms such as Fido, Carbon, Branch, and numerous smaller operators now constitute a significant, if lightly regulated, segment of Ghana’s consumer finance market.

The value proposition is compelling for a population with limited access to traditional bank credit. No collateral is required. No physical branch visit is necessary. The application process takes minutes, and funds are disbursed instantly to the borrower’s MoMo wallet. Repayment is equally frictionless, deducted automatically from the same wallet on the due date. For the financially excluded—the informal sector worker, the market trader, the student—digital lending represents access to credit that the formal banking sector has historically denied (21).

However, the same structural characteristics that make digital lending accessible also make it dangerous. Mogaji and Nguyen (1) flagged the debt-cycle risk early: FinTech developer informants in their study candidly acknowledged that consumers are “tempted to borrow money they might not be able to repay,” creating a cycle of indebtedness facilitated by the very frictionlessness that makes the product attractive. The integration with MoMo infrastructure means that a loan can be taken, spent, and defaulted upon within the same ecosystem that holds the borrower’s primary financial identity.

The digital lending sector in Ghana thus presents a threat landscape that is distinct from, but deeply connected to, the broader MoMo ecosystem. It is a double threat: the platform is simultaneously a victim of sophisticated fraud perpetrated by external actors, and a perpetrator of data weaponization against its own customers. This chapter examines both dimensions.

4.2 Threat One: The Platform as Victim

Digital lending platforms, like all credit providers, face the fundamental challenge of determining which applicants will repay and which will default. Unlike traditional banks, which can rely on lengthy application processes, collateral requirements, and relationship-based lending, digital lenders must make credit decisions algorithmically, often within minutes, based on limited data (15). This creates specific vulnerabilities that fraudsters systematically exploit.

4.2.1 Synthetic Identity Fraud

Synthetic identity fraud—the combination of genuine and fabricated personal information to create a new, fictitious identity—is the most structurally damaging fraud type facing digital lenders. A fraudster combines a legitimate Bank Verification Number (BVN) or Ghana Card number with fabricated biographic details, creating an identity that passes automated KYC checks but has no real human behind it. This synthetic identity then applies for loans across multiple platforms, maximising the total credit obtained before the identity is abandoned.

The technical sophistication of synthetic identity generation is increasing rapidly. Bello et al. (20) document that deepfake and synthetic identity threats are now empirically confirmed as bypassing biometric authentication systems. Arthur et al. (3) confirm that AI-enabled fraud—including synthetic identity generation—accounted for 47.2% of cybercrime cases in Ghana by the first half of 2025, with losses 2.73 times higher than non-AI cases. The fraudster’s toolkit now includes generative adversarial networks capable of producing realistic Ghana Card images and supporting documentation that automated verification systems cannot reliably distinguish from genuine documents.

4.2.2 Loan Stacking

Loan stacking—the practice of taking multiple loans from different platforms simultaneously with no intention of repayment—exploits a structural weakness in the digital lending market: the absence of a shared credit bureau that provides real-time visibility across all lenders. A fraudster with a synthetic identity, or even a genuine borrower who has decided to maximise their borrowing before defaulting, can apply to ten or more platforms in rapid succession. Each platform’s algorithm evaluates the application in isolation, unaware that the applicant has just been approved for loans elsewhere.

The speed of the MoMo infrastructure amplifies this vulnerability. Once approved, funds are disbursed instantly to the borrower’s wallet and can be transferred or cashed out within minutes. By the time one lender reports a default to a credit bureau—a process that may take days or weeks—the fraudster has already monetised loans from multiple platforms and abandoned the identity.

4.2.3 Credential Stuffing and Account Takeover

Credential stuffing—the automated use of leaked username and password combinations from data breaches at other services to gain unauthorised access to accounts—is a threat that digital lenders share with the broader financial ecosystem. A borrower who uses the same password for their loan app as they used for a social media account that was breached in an unrelated incident is vulnerable to account takeover. Once inside the account, the attacker can change the linked MoMo wallet, apply for additional loans, and redirect disbursements to accounts under their control.

This threat is particularly acute for digital lending platforms because their customer base overlaps significantly with the population segments that Arthur et al. (3) identify as having lower digital literacy and, consequently, poorer password hygiene. The same users who are most in need of accessible credit are also the users most likely to reuse passwords across services.

4.2.4 Referral Program Abuse

Many digital lending platforms use referral programs as a customer acquisition strategy: existing customers receive a bonus or reduced interest rate for each new customer they refer. These programs are trivially exploitable. A fraudster creates multiple synthetic identities, uses each to refer the others, and collects the referral bonuses without any genuine lending activity taking place. The financial loss per incident is typically small, but the volume can be significant, and the operational cost of manually investigating each suspicious referral pattern can exceed the direct financial loss.

4.3 Threat Two: The Platform as Predator

If the threats described above represent the digital lending platform as victim, the following analysis examines the platform as perpetrator. The same data-driven, frictionless model that enables rapid credit decisions also enables practices that range from ethically questionable to clearly illegal under Ghana’s Data Protection Act, 2012 (Act 843).

4.3.1 Excessive Permission Harvesting

When a user installs a digital lending application, they are typically presented with a list of permissions the app requires. Many of these permissions have no functional relationship to credit assessment. Contact lists. SMS logs. Call logs. Camera. Location. Photo gallery. The app requests access to the most intimate corners of the user’s digital life.

The justification offered by lenders is that this data enables more accurate credit scoring—analysing a borrower’s contact list, for example, might reveal the creditworthiness of their social network. Upadhyaya et al. (16), in their study of Kenya’s digital credit

market, found that this justification masks a darker purpose. The data is not used solely, or even primarily, for credit assessment. It is used for debt collection.

4.3.2 The Debt-Shaming Playbook

The debt-shaming playbook, documented extensively in the Kenyan context by Upadhyaya et al. (16) and increasingly reported in Ghanaian media, follows a predictable and brutal sequence.

When a borrower misses a repayment deadline, the lender's first step is automated reminders via SMS and in-app notifications. If the borrower does not respond, the escalation begins. The lender, having harvested the borrower's contact list at the point of installation, begins contacting the borrower's family members, friends, colleagues, and employers. The messages inform these third parties that the borrower has defaulted on a loan and urge them to pressure the borrower to repay.

In more aggressive cases, the lender creates WhatsApp groups containing the borrower and multiple contacts from their phone book, broadcasting the borrower's indebtedness to their entire social network. The psychological impact of this practice—being publicly shamed before family, friends, professional contacts, and community members—is devastating. Upadhyaya et al. (16) cite 2019 FinAccess survey data showing that 51% of Kenyan digital loan borrowers reported having to sell assets, borrow from family, or reduce food expenditure to repay digital loans. The debt-shaming mechanism is a significant driver of this distress.

The Ghanaian regulatory framework explicitly prohibits these practices. The Data Protection Act, 2012 (Act 843) establishes principles of data minimisation, purpose limitation, and explicit consent (24). Harvesting a borrower's contact list for the stated purpose of credit assessment and then using that same data for debt shaming is a clear violation of the purpose limitation principle. Contacting third parties to disclose a borrower's indebtedness violates the data subject's right to privacy. Yet, as Zibo and Nweke (24) document, enforcement of Act 843 remains weak. The Data Protection Commission is under-resourced, and public awareness of data protection rights is low.

4.3.3 Data Exploitation Beyond Shaming

The data harvested by digital lending apps has value beyond debt collection. Contact lists can be sold to marketing companies. SMS logs, which often contain transaction alerts from MoMo and banking apps, can be mined for intelligence about a user's financial behaviour—intelligence that is valuable to competitors, fraudsters, or political actors (24). The borrower, having clicked "Accept" on a permissions screen they likely did not read or fully understand, has surrendered control of their personal data to an entity with minimal regulatory oversight.

The economic incentives are perverse. A digital lender that spends heavily on KYC and data protection compliance is at a competitive disadvantage relative to a lender

that cuts corners, harvests data aggressively, and uses that data to maximise recoveries through social pressure. The regulatory vacuum identified by Upadhyaya et al. (16) in Kenya—where regulation targeted unregulated non-bank lenders while leaving the dominant bank–MNO partnership products largely untouched—is directly relevant to Ghana. If Ghana’s regulatory response to digital lending follows the Kenyan pattern, the most aggressive data harvesters will continue to operate with impunity while the formal banking sector, which is subject to BoG oversight, bears the compliance cost.

4.4 The Convergence: A Unified Criminal Ecosystem

The digital lending threat landscape cannot be understood in isolation from the broader MoMo and social engineering ecosystems analysed in Chapters 2 and 3. These are not separate threat domains. They are interconnected nodes in a unified criminal economy.

A SIM swap attack (Chapter 2, Section 2.3.1) that compromises a victim’s MoMo wallet simultaneously compromises any digital lending accounts linked to that wallet. The fraudster can apply for loans in the victim’s name and redirect disbursements to a controlled account. The victim discovers the fraud not only when their MoMo balance disappears but when they begin receiving debt-shaming calls for loans they never took.

A vishing attack (Chapter 3, Section 3.2.2) that extracts a victim’s personal information—date of birth, Ghana Card number, mother’s maiden name—provides the raw material for synthetic identity creation. The fraudster who obtains this data through social engineering can use it to apply for digital loans, and the data trail will point not to the fraudster but to the vishing victim.

A digital lending platform that harvests contact lists and SMS logs accumulates a database of phone numbers, financial behaviours, and social relationships that is, from a social engineer’s perspective, a targeting list. The fraudster who acquires this data—whether by purchasing it from a corrupt lender employee, extracting it through a data breach, or simply operating their own lending platform as a data-harvesting front—can craft highly personalised smishing and vishing campaigns against individuals whose financial vulnerability has already been profiled.

This convergence is the most insidious dimension of the digital lending threat landscape. The platform that presents itself as a legitimate credit provider may, in reality, be a data-harvesting operation whose primary revenue stream is not loan interest but the monetisation of borrower data through debt shaming, sale to third parties, and enablement of downstream fraud. The victim, having consented to a loan agreement they did not fully understand, becomes both a debtor and a data product.

4.5 The Regulatory Blind Spot: Lessons from Kenya

Ghana’s digital lending sector is younger than Kenya’s, and the regulatory response is correspondingly less developed. The Kenyan experience, documented in detail by

Upadhyaya et al. (16), offers a cautionary case study with direct structural relevance.

Kenya's Central Bank of Kenya (Amendment) Act 2021 brought digital lenders under central bank licensing for the first time. The intervention was well-intentioned but structurally limited: it almost exclusively targeted previously unregulated non-bank lenders, while leaving the dominant bank–Safaricom partnership products—Fuliza, M-Shwari, KCB M-Pesa, which together controlled over 80% of the digital credit market—largely untouched. The regulation changed the behaviour of the smallest, least systemically important actors while leaving the market leaders free to continue practices that included effective annual percentage rates as high as 442% and the debt-shaming tactics described above.

Upadhyaya et al. (16) explain this outcome through the lens of “structural power” rather than simple regulatory capture. Safaricom's economic centrality—the company pays approximately 10% of Kenya's total corporate tax—gave the state a strong incentive not to over-regulate its digital lending products. The regulatory outcome was not the result of corruption or lobbying in the conventional sense, but of the structural reality that a regulator cannot easily impose costs on an entity whose financial health is material to the national economy.

The Ghanaian parallel is clear. MTN Ghana is to Ghana's mobile money ecosystem what Safaricom is to Kenya's. Any digital lending regulation that exempts, or lightly regulates, the bank–MNO partnership products that dominate the market will replicate the Kenyan outcome: regulatory appearance without consumer protection substance. This lesson will inform the recommendations in Chapter 6.

4.6 Chapter Summary

The digital lending sector in Ghana represents a frontier where the line between victim and predator is dangerously blurred. The platforms are genuine victims of sophisticated fraud—synthetic identities, loan stacking, credential stuffing, and referral program abuse—that exploits the algorithmic, high-speed nature of their credit decisioning. But they are also perpetrators of data weaponization, harvesting contact lists and SMS logs under the guise of credit assessment and deploying that data for debt shaming, third-party disclosure, and commercial exploitation that violates the letter and spirit of Ghana's Data Protection Act.

The convergence of digital lending with the broader MoMo and social engineering ecosystems creates a unified criminal economy in which a single SIM swap can compromise a victim's entire mobile financial identity, including loans they never applied for, while the data harvested by lending platforms provides the targeting intelligence for the next generation of social engineering attacks. The Kenyan regulatory experience demonstrates that partial regulation—targeting standalone fintech lenders while exempting bank–MNO partnerships—is worse than no regulation at all, creating a two-tier system in which the most systemically important actors face the least oversight.

This is the predatory frontier. The regulatory framework that has failed to contain it is the subject of the next chapter.

Chapter 5

The Framework: The Fragmented Regulatory Response

5.1 The Regulatory Triangulation Problem

If the Mobile Money ecosystem is the battlefield and social engineering is the weapon, the regulatory framework is the shield—and it is a shield with three handles held by four different hands, none of which has a clear mandate to coordinate with the others.

Ghana’s mobile financial security is governed by a distributed architecture of regulatory bodies whose jurisdictions overlap, conflict, and leave gaps that threat actors systematically exploit. The Bank of Ghana (BoG) regulates the financial system, including the partner banks that hold MoMo trust accounts and the payment systems that process transactions. The National Communications Authority (NCA) regulates the telecommunications infrastructure—the SIM cards, the USSD channels, the mobile networks—upon which the entire MoMo ecosystem depends. The Data Protection Commission (DPC) is responsible for enforcing the Data Protection Act, 2012 (Act 843), which governs the collection, processing, and protection of personal data. And the Cyber Security Authority (CSA), established by the Cybersecurity Act, 2020 (Act 1038), holds the broad mandate for national cybersecurity coordination, including the protection of Critical Information Infrastructure (CII) into which category the financial sector falls.

This distributed architecture creates what can be termed a “regulatory triangulation problem.” A SIM swap fraud that drains a victim’s MoMo wallet implicates all four bodies simultaneously: the NCA, because the fraud exploited a weakness in SIM registration and porting processes; the BoG, because the financial loss occurred through a regulated payment system; the DPC, because the victim’s personal data was used to authenticate the fraudulent SIM swap; and the CSA, because the incident is a cybercrime affecting Critical Information Infrastructure. Yet no single body has clear, primary jurisdiction. The victim does not know where to complain. The institutions themselves lack a formal coordination mechanism for real-time incident response (3; 14).

Nartey et al. (14) provide the most comprehensive analysis of Ghana’s cybersecurity governance architecture. Ghana has ratified both the Malabo Convention and the

Budapest Convention, placing it within the framework of international cybercrime co-operation. The country ranked third in Africa on the International Telecommunication Union’s Global Cybersecurity Index in 2020, reflecting significant investment in legal and institutional capacity. Yet the same study notes that as of 2024, only four of twelve Economic Community of West African States (ECOWAS) member states had ratified the Malabo Convention, and 62% of surveyed councils across five ECOWAS states had no cybersecurity budget. Fewer than 5% had an incident response plan. Ghana’s regional leadership, while genuine, exists in a neighbourhood of limited capacity—and cross-border fraud does not respect the capacity gradient.

5.2 The Liability Vacuum

The most immediate human consequence of the regulatory triangulation problem is the liability vacuum. When a Ghanaian MoMo user is defrauded through a SIM swap, who bears the financial loss?

The telecommunications provider, whose agent issued a replacement SIM without adequate verification, bears operational responsibility for the SIM swap event. But the MNO is regulated by the NCA, not the BoG, and the NCA’s mandate does not extend to compensating victims of financial fraud. The bank that held the linked account bears responsibility for the security of its systems, but the fraud did not originate in the bank’s infrastructure—it originated at the telecom agent’s point of service. The BoG’s Cyber and Information Security Directive mandates that financial institutions report breaches within 24 hours and maintain adequate security controls, but the directive does not establish a clear liability framework for losses incurred at the intersection of telecom and banking infrastructure (17).

The result, documented repeatedly in Ghanaian media reports and victim accounts, is a blame game in which each institution directs the victim to another. The telecom provider tells the victim to contact their bank. The bank tells the victim the fraud originated with the telecom provider. The regulator receives a complaint but lacks the statutory authority to order compensation. The victim, already financially vulnerable, absorbs the loss.

This liability vacuum is not merely an administrative inconvenience. It is a structural threat to the trust upon which the entire digital financial ecosystem depends. Arthur et al. (3) found that 47.3% of rural MoMo users who experienced fraud subsequently reduced or stopped using mobile money services, compared to 18.9% of urban users. The liability vacuum does not just harm individual victims; it erodes the financial inclusion gains that MoMo was designed to achieve. The Digital Inclusion Paradox thus acquires a regulatory dimension: the framework that was supposed to protect users is so fragmented that it accelerates the very exclusion it was built to prevent.

5.3 The Cybersecurity Act, 2020 (Act 1038): Ambition and Implementation

The Cybersecurity Act, 2020 (Act 1038) is Ghana’s flagship cybersecurity legislation. It established the Cyber Security Authority (CSA), designated Critical Information Infrastructure (CII) sectors—including financial services—and created a framework for incident reporting, capacity building, and international cooperation. On paper, it is a comprehensive and forward-looking piece of legislation (14).

In practice, its implementation has been constrained by three factors: statutory gaps, resource limitations, and what Egbobamwonyi-Bedaux (22) terms “digital dependency”—the structural reality that Ghana’s cybersecurity infrastructure is shaped less by sovereign policy choice than by dependence on foreign technology, standards, and threat classifications.

5.3.1 Statutory Gaps: The AI Blind Spot

The most significant statutory gap in Act 1038, from the perspective of the threat landscape analysed in this report, is its silence on artificial intelligence. Mensah et al. (18) conducted a detailed legal assessment of Act 1038, examining Sections 5, 32(1), 56, and 57. Their finding is unambiguous: the Act does not explicitly mention AI systems, machine learning vulnerabilities, or AI-enabled threats in its capacity-building, incident reporting, or CII designation provisions.

Section 32(1) defines Critical Information Infrastructure broadly enough that it could, in principle, encompass AI-driven financial decision systems—but only if the CSA proactively designates them as such. The CSA has not done so. Section 56 and 57 address training and capacity building, covering forensics, vulnerability detection, and incident reporting, but do not directly address AI-specific vulnerabilities: data poisoning, adversarial attacks on machine learning models, model theft, algorithmic bias, or—most critically for the threats documented in this report—deepfake-enabled social engineering and synthetic identity fraud.

Arthur et al. (3) reinforce this critique from the empirical side. Their finding that AI-enabled elements were present in 47.2% of cybercrime cases in Ghana by H1 2025, with losses 2.73 times higher than non-AI cases, demonstrates that the statutory gap identified by Mensah et al. (18) is not a theoretical future concern. It is a present operational reality. Ghana’s primary cybersecurity law does not address the technology that is driving the fastest-growing and most damaging category of financial cybercrime.

5.3.2 Resource Constraints and Enforcement Capacity

A law is only as effective as the institution that enforces it. Opuni-Frimpong et al. (17) provide the most detailed Ghana-specific empirical evidence on this point through their analysis of the Bank of Ghana’s Cyber and Information Security Governance Committee

(CISGC) directive.

The BoG’s CISGC directive, issued in 2018 and made mandatory for universal banks from 2019, requires every regulated bank to establish a board-level cybersecurity committee. Opuni-Frimpong et al. analysed data from 20 Ghanaian universal banks over the period 2019–2022 and found that only 19.3% of CISGC members, on average, possess cyber or IT expertise. The study found that CISGC cyber expertise has a statistically significant positive impact on return on assets—expertise matters—but committee size, meeting frequency, and female representation showed no significant impact on financial performance or efficiency. In other words, the BoG’s directive created committees, but did not ensure that those committees were populated with members who understand the threats they are supposed to govern.

The resource constraint extends beyond the private sector to the public enforcement bodies. Nartey et al. (14) note chronic under-resourcing of the Ghana Police Service’s Cybercrime Unit outside the Accra and Cape Coast metropolitan areas. A victim of MoMo fraud in a rural district in the Northern Region or the Volta Region faces not only the liability vacuum described above but also a practical enforcement vacuum: the nearest cybercrime investigator may be hundreds of kilometres away, and the cost of pursuing the case exceeds the value of the loss.

5.3.3 Digital Dependency and Imported Frameworks

Egbobamwonyi-Bedaux (22) offers a more fundamental critique of Ghana’s cybersecurity governance, grounded in dependency theory and decolonial analysis. The argument, applied to the financial sector, is that Ghana’s cybersecurity priorities are shaped less by an autonomous assessment of local threats than by dependence on foreign infrastructure, cloud providers, pre-trained AI models, and internationally circulated “best practice” frameworks.

The practical consequence is that externally developed threat classifications tend to prioritise globally visible risks—ransomware, cross-border espionage, GDPR-style privacy violations—while locally salient threats such as MoMo-specific social engineering, SIM swap fraud, and dialect-based phishing receive limited analytical visibility in the international frameworks that shape donor funding, capacity-building programs, and regulatory benchmarking. International indices like the ITU Global Cybersecurity Index, on which Ghana performs well, “responsibilise” Ghanaian agencies to conform to imported metrics of progress rather than responsiveness to the local threat landscape (22).

This critique does not diminish the value of Ghana’s international cooperation or its legislative achievements. It does, however, explain a pattern that is visible throughout the threat landscape analysed in this report: Ghana has world-class cybersecurity legislation on paper, but the operational capacity to enforce it against the specific threats that most affect Ghanaian citizens remains underdeveloped. The gap is not in the law’s ambition but in its grounding in local threat reality.

5.4 The Data Protection Act, 2012 (Act 843): Enforcement Without Teeth

The Data Protection Act, 2012 (Act 843) is Ghana’s foundational personal data protection legislation. It establishes principles of data minimisation, purpose limitation, explicit consent, and data subject rights that are broadly consistent with international standards, including the European Union’s General Data Protection Regulation (GDPR) (24).

The Act’s relevance to the threat landscape analysed in this report is direct and specific. The debt-shaming practices documented in Chapter 4—harvesting a borrower’s contact list under the guise of credit assessment and then using that data to contact family, friends, and employers to disclose indebtedness—violate multiple provisions of Act 843. The initial harvesting of contact data for a purpose (credit assessment) that differs from the actual use (debt collection) violates the purpose limitation principle. The disclosure of a borrower’s indebtedness to third parties without consent violates the data subject’s right to privacy. The absence of meaningful consent—users click “Accept” on a permissions screen they did not read or understand—violates the requirement for informed, explicit consent.

Yet enforcement is minimal. Zibo and Nweke (24) document persistent enforcement and awareness gaps in Act 843’s implementation. The Data Protection Commission, while established and operational, lacks the resources and political authority to pursue aggressive enforcement actions against powerful commercial actors. Public awareness of data protection rights is low, particularly among the financially vulnerable populations most likely to use digital lending platforms. The result is a data protection law that exists on paper but offers little practical protection to the citizens whose data is harvested, monetised, and weaponised.

The comparative analysis provided by Zibo and Nweke is instructive. When Act 843 is measured against the GDPR and the California Consumer Privacy Act (CCPA), it compares favourably in scope—it applies to all data controllers and processors operating in Ghana. It compares unfavourably in enforcement: the penalties available under Act 843 are weaker, the enforcement body is less resourced, and the public awareness that drives complaints and compliance is lower. A digital lending platform that would face significant regulatory risk for debt-shaming practices in the European Union faces minimal risk in Ghana, despite the legal prohibition being substantively similar.

5.5 The Missing Mandates

Table 5.1: Ghana’s Mobile Financial Security Regulatory Framework: Mandated vs. Missing Requirements

Domain	What the Law Requires	What Is Missing
Cybersecurity Governance	Act 1038 designates financial services as Critical Information Infrastructure (CII). The BoG Cyber and Information Security Directive mandates board-level cybersecurity committees (CISGC) at all universal banks (17; 14).	Only 19.3% of CISGC members possess cyber/IT expertise. Act 1038 contains no explicit provisions addressing AI-enabled threats such as deepfakes, synthetic identity fraud, or adversarial machine learning attacks (18).
Data Protection	Act 843 establishes principles of data minimisation, purpose limitation, explicit consent, and data subject rights. Applies to all data controllers and processors operating in Ghana (24).	No sector-specific regulation governs which app permissions digital lending platforms may request. Debt-shaming practices that weaponise borrower contact lists continue with minimal enforcement. DPC lacks resources for aggressive enforcement against major commercial actors (24).
Incident Reporting	BoG Cyber and Information Security Directive mandates 24-hour breach notification for regulated financial institutions (17).	No cross-sector incident response framework exists. A SIM swap fraud implicating an MNO, a bank, and a data breach has no single coordinating authority. The Police Cybercrime Unit is chronically under-resourced outside Accra and Cape Coast (14).
Security Awareness Training	Financial institutions are required to implement security controls under the BoG directive. General staff training obligations exist under broader corporate governance requirements.	No mandate for psychologically informed, culturally tailored security awareness training. No requirement for simulated social engineering campaigns. The MoMo agent network—over 300,000 agents—has no mandatory cybersecurity training or certification requirement (19).
Digital Lending Regulation	The BoG regulates financial institutions. Act 843 applies to data processing by all entities.	No binding directive specifically governs digital lending data ethics. No whitelist of permissible app permissions for credit assessment. No regulatory coverage of bank–MNO partnership lending products, which dominate market share. The Kenyan experience demonstrates that partial regulation excluding dominant telco-bank products leaves consumers unprotected (16).

Beyond the gaps within existing legislation, the regulatory framework is defined as much by what it does not require as by what it does. Three missing mandates are particularly consequential for the threat landscape analysed in this report.

5.5.1 No Mandated Security Awareness Training

The Bank of Ghana’s Cyber and Information Security Directive requires financial institutions to implement security controls and report incidents. It does not mandate specific, psychologically informed security awareness training for staff or for the MoMo agent network. The distinction is critical. Generic, annual, computer-based training modules—the industry default—have limited effectiveness against the culturally tailored, linguistically specific social engineering attacks documented in Chapter 3 (19). Adewopo et al. (19) argue for a “culture-centered” prevention approach that leverages local cultural values and ethics, but no Ghanaian regulatory instrument requires financial institutions to adopt such an approach.

The agent network, which Chapter 2 identified as the most significant human vulnerability in the MoMo ecosystem, is entirely outside the scope of any mandatory cybersecurity training regime. Agents are trained in customer service and transaction processing, not in social engineering detection or fraud prevention. Yet they are the human endpoints with the authority to authorise cash-out transactions and process KYC verifications—the very functions that fraudsters exploit.

5.5.2 No Data Permission Rules for Digital Lending Applications

Act 843 establishes general data protection principles, but no sector-specific regulation governs the data permissions that digital lending applications may request at the point of installation. An app that requests access to a user’s contact list, SMS logs, call logs, camera, and location is not violating any specific regulatory prohibition—even though, as Chapter 4 demonstrated, that data is routinely used for purposes far beyond legitimate credit assessment.

The Kenyan experience analysed by Upadhyaya et al. (16) demonstrates that bringing digital lenders under central bank licensing is not, by itself, sufficient. The Kenyan regulation did not impose specific data permission restrictions, and debt-shaming practices have continued. A binding directive from the Data Protection Commission, or a joint directive from the DPC and the BoG, that specifies which app permissions are permissible for credit assessment purposes and which are prohibited, would close this gap. No such directive currently exists.

5.5.3 No Cross-Sector Incident Response Framework

A SIM swap fraud that drains a MoMo wallet and compromises linked digital lending accounts is a single incident with implications for telecommunications regulation, banking regulation, data protection, and criminal law enforcement. Responding to it effectively

requires coordination across the NCA, the BoG, the DPC, and the Ghana Police Service. No formal framework exists for such coordination.

The INTERPOL Operation Contender 2.0 case documented by Nartey et al. (14)—in which eight cybercriminals behind an Abidjan-based phishing operation were arrested, with over 260 victim reports in Switzerland and confirmed losses over US\$1.4 million—demonstrates that cross-border coordination is possible when international law enforcement takes the lead. The same level of coordination has not been institutionalised for domestic incidents within Ghana’s own regulatory architecture.

5.6 Chapter Summary

Ghana’s regulatory framework for mobile financial security is comprehensive on paper and fragmented in practice. The Cybersecurity Act 2020 established a world-class legislative foundation but contains no provisions addressing AI-enabled threats that now account for nearly half of all cybercrime cases. The Data Protection Act 2012 prohibits the data weaponization practices that characterise the digital lending sector but lacks the enforcement capacity to deter them. The Bank of Ghana’s cybersecurity governance directive has created board-level committees that lack cyber expertise. And the distributed architecture of the BoG, NCA, DPC, and CSA creates a liability vacuum in which victims of the most common form of mobile financial fraud—SIM swap-enabled MoMo theft—have no clear path to restitution.

The regulatory framework, like the MoMo platform it governs, was built for a different threat landscape than the one that now exists. It assumes clear jurisdictional boundaries that cybercriminals ignore. It assumes a pace of technological change that AI has rendered obsolete. And it assumes a level of institutional capacity that resource constraints have prevented from materialising.

The shield is broken. The next chapter proposes how to fix it.

Chapter 6

Strategic Recommendations: Building a Human-Centric Defence

The preceding chapters have established that the greatest cyber threat to Ghana’s financial sector is not a piece of code but the systematic manipulation of human beings, enabled by vulnerable mobile platforms, exploited by predatory digital lending models, and allowed to fester in a fragmented regulatory landscape. This chapter translates that analysis into action. Five strategic recommendations are presented, prioritised by impact and feasibility, each grounded in the evidence assembled throughout this report.

6.1 Recommendation 1: Establish a National Mobile Financial Fraud Coordination Framework (NM3F)

The regulatory triangulation problem documented in Chapter 5—the distribution of authority across the Bank of Ghana, the National Communications Authority, the Data Protection Commission, and the Cyber Security Authority without a formal coordination mechanism—is the single most consequential structural weakness in Ghana’s mobile financial security architecture. It must be addressed first, because all other recommendations depend on effective inter-agency coordination for their implementation.

Proposed Action: The Ministry of Communications and Digitalisation, in conjunction with the Ministry of Finance, should establish a National Mobile Financial Fraud Coordination Framework (NM3F). The NM3F would be a joint operational task force, not merely a policy coordination committee, with the following characteristics:

- **Permanent co-staffing:** The BoG, NCA, DPC, and CSA would each second personnel to the NM3F on a rotating basis, ensuring that institutional knowledge is retained and that no single agency’s perspective dominates.
- **Real-time intelligence sharing:** The NM3F would operate a shared threat intelligence platform, accessible to all member agencies and, on a read-only basis with appropriate confidentiality controls, to regulated financial institutions and Mobile Network

Operators. Phishing URLs, fraudulent phone numbers, compromised agent identities, and emerging attack patterns would be shared within minutes, not weeks.

- **Coordinated takedown capability:** The NM3F would have the operational authority to coordinate the takedown of phishing infrastructure across telecommunications networks, financial platforms, and internet service providers, eliminating the current situation in which a phishing site may be reported to one agency but remain active on infrastructure regulated by another.
- **Unified public awareness messaging:** The NM3F would serve as the single authoritative source of public-facing fraud alerts, replacing the current fragmented landscape in which individual banks, MNOs, and agencies issue separate warnings that may conflict or duplicate each other.

Evidence Base: The INTERPOL Operation Contender 2.0 case, documented by Nartey et al. (14), demonstrates the operational effectiveness of coordinated, multi-jurisdictional fraud response. The NM3F would institutionalise this coordination domestically. The Italy cross-industry API model, described by Kim et al. (4), in which banks query telecommunications providers for recent SIM-change history before approving high-risk transactions, provides a concrete technical template that the NM3F could adapt for the Ghanaian context. The “structural power” analysis of Upadhyaya et al. (16) underscores the importance of designing the NM3F with sufficient independence that it is not captured by the dominant market actors it must regulate.

6.2 Recommendation 2: Mandate Psychological Inoculation Training Across the Financial Sector

The social engineering taxonomy presented in Chapter 3 demonstrates that Ghanaian mobile financial fraud is culturally tailored, linguistically specific, and psychologically sophisticated. The countermeasure must be equally sophisticated. Generic, annual, computer-based training modules—the current industry default—are ineffective against attacks that exploit local languages, communal trust norms, and culturally specific authority relationships (19).

Proposed Action: The Bank of Ghana, in consultation with the Cyber Security Authority, should amend the Cyber and Information Security Directive to mandate:

- **Frequent, simulated social engineering campaigns:** Regulated financial institutions would be required to conduct simulated smishing, vishing, and WhatsApp impersonation campaigns against their own staff and customers at least quarterly, using localised scripts in Twi, Ewe, Ga, Hausa, and Ghanaian Pidgin English. The campaigns would be psychologically informed, incorporating Cialdini’s six principles of persuasion—authority, scarcity, reciprocity, commitment and consistency, social proof,

and liking—that Karadsheh et al. (10) empirically validate as the psychological foundation of social engineering success.

- **Extension to the agent network:** The training mandate would extend to the entire MoMo agent network. Agents would be required to complete a certified fraud detection module as a condition of licence renewal. The module would cover recognition of common social engineering scripts, verification of customer identity beyond SIM possession, and escalation procedures for suspected fraud. The Bank of Ghana’s reported blacklisting of 1,200 agents in 2024 (2) demonstrates both the scale of the problem and the existing regulatory capacity for agent-level enforcement—capacity that should be expanded from punitive action to preventive training.
- **Culture-centered design:** Training content would be developed in consultation with Ghanaian psychologists, linguists, and community leaders, following the “culture-centered” prevention model proposed by Adewopo et al. (19). The content would leverage local cultural values and ethics—communal responsibility, religious integrity, respect earned through protection of the vulnerable—rather than importing Western individualistic “trust no one” messaging that is culturally alien in a relational trust society (9).

Evidence Base: The empirically validated attack-phase model of Karadsheh et al. (10) demonstrates that “Social Engineer Involvement”—the skill with which the attacker executes each phase—is the single strongest predictor of attack success ($\beta = 0.38$). Training that inoculates potential victims against skilled social engineering execution directly targets this vulnerability. The MTN Ghana “Stay Safe” SMS awareness campaign, which reportedly improved vishing recognition by 50% among 8 million subscribers (2), provides a proof of concept for the effectiveness of MNO-led awareness initiatives, though these claims require verification against primary MTN Ghana sources.

6.3 Recommendation 3: Issue a Binding Digital Lending Data Ethics Directive

Chapter 4 documented the digital lending sector’s double threat: platforms are simultaneously victims of fraud and perpetrators of data weaponization against consumers. The debt-shaming playbook—harvesting contact lists under the guise of credit assessment and deploying that data for public humiliation of defaulters—violates the letter and spirit of the Data Protection Act, 2012 (Act 843) (24). Yet enforcement is minimal, and the practice continues.

Proposed Action: The Data Protection Commission, in coordination with the Bank of Ghana, should issue a binding Digital Lending Data Ethics Directive with the following provisions:

- **Explicit prohibition of contact list harvesting for debt collection:** The directive would state unambiguously that personal data collected for credit assessment

purposes may not be used for debt collection, and that contacting third parties to disclose a borrower’s indebtedness without the borrower’s explicit, specific consent is a violation of the Data Protection Act subject to administrative penalties.

- **App permission whitelist:** The directive would specify a whitelist of app permissions that digital lending applications may request at the point of installation. Contact lists, SMS logs, call logs, and photo galleries—data categories with no legitimate, proportionate relationship to credit assessment—would be excluded from the whitelist. Applications requesting permissions outside the whitelist would be denied access to the Ghanaian market.
- **App store engagement:** The DPC and BoG would formally engage Google and Apple to establish a process for delisting digital lending applications that violate the directive, following the model of Kenya’s 2022 Google Play Store policy change that removed non-compliant loan apps.
- **Regulatory scope covering bank–MNO partnership products:** The directive would explicitly apply to digital lending products offered through bank–MNO partnerships (such as MTN MoMo-linked loan products), not solely to standalone fintech lenders. The Kenyan experience documented by Upadhyaya et al. (16) demonstrates that regulation targeting only non-bank lenders while exempting the dominant bank–MNO partnership products leaves the majority of the market unregulated and consumers unprotected.
- **Penalties with deterrent effect:** The directive would specify administrative fines calibrated to the revenue of the infringing entity, ensuring that the penalty for non-compliance exceeds the profit derived from the infringing practice. Licence revocation would be available as a sanction of last resort for repeated or egregious violations.

Evidence Base: The Kenyan case study of Upadhyaya et al. (16) provides the primary comparative evidence. Kenya’s regulation brought digital lenders under central bank licensing but failed to impose specific data ethics requirements, and debt-shaming practices continued. The Ghanaian directive proposed here would learn from that failure by regulating practices, not merely licensing entities. The structural power analysis of the same study—the finding that Safaricom’s economic centrality limited regulatory ambition—is directly transferable to Ghana and underscores the importance of ensuring that the directive covers MTN-linked products from inception, before a pattern of non-enforcement becomes entrenched.

6.4 Recommendation 4: Fix the Liability Architecture

The liability vacuum documented in Chapter 5—the absence of a clear, time-bound framework for determining who bears the financial loss when a MoMo user is defrauded through SIM swap or social engineering—is a structural injustice that erodes trust in the entire

digital financial ecosystem. A victim who loses their savings to a SIM swap and is then passed between their telecom provider, their bank, and multiple regulators without restitution is a victim of regulatory failure as much as of criminal fraud.

Proposed Action: The Bank of Ghana and the National Communications Authority should issue a joint directive establishing a clear, tiered liability framework for mobile financial fraud:

- **Tier 1 — MNO liability for SIM swap fraud:** Where a fraud is enabled by an unauthorised SIM swap or SIM porting event, the telecommunications provider whose agent or system processed the fraudulent swap bears primary liability for the financial loss. This allocation reflects the operational reality that the fraud was enabled by a failure in the MNO’s customer authentication process, not by any action of the victim or the victim’s bank.
- **Tier 2 — Bank liability for platform security failures:** Where a fraud is enabled by a failure in a bank’s or fintech’s platform security—including inadequate transaction monitoring, failure to implement two-factor authentication meeting NIST AAL2 standards as recommended by Lamoyero and Fajana (6), or failure to verify SIM-change history before approving high-risk transactions—the financial institution bears primary liability.
- **Tier 3 — Shared liability for complex fraud:** Where a fraud involves failures by both an MNO and a financial institution, liability is apportioned according to the relative contribution of each party’s failure, as determined by the NM3F coordination body proposed in Recommendation 1.
- **Time-bound resolution:** The directive would establish a maximum resolution period—30 calendar days from the date of the victim’s complaint—within which the liable party must compensate the victim. Failure to resolve within this period would trigger automatic escalation to the BoG and NCA for regulatory enforcement.
- **Victim contributory negligence:** The directive would define a clear standard for victim contributory negligence—for example, cases in which the victim voluntarily shared their PIN with a third party in response to an obvious phishing attempt—that would limit or extinguish the victim’s right to compensation. This provision protects the system against moral hazard while ensuring that victims who took reasonable care are protected.

Evidence Base: The finding of Arthur et al. (3) that 47.3% of rural fraud victims subsequently reduce or stop using MoMo services demonstrates the systemic cost of the liability vacuum. Each uncompensated victim represents not only an individual injustice but a threat to the financial inclusion gains that MoMo was designed to achieve. The Italy cross-industry API model described by Kim et al. (4) provides a technical template for the SIM-change verification that would support Tier 1 liability determinations.

6.5 Recommendation 5: Launch a National “Verify Before You Trust” Public Education Campaign

The psychological analysis presented in Chapter 3 demonstrates that Ghanaian social engineering victims are not uniquely careless or technologically incompetent. They are targeted by attacks that are specifically designed to exploit Ghanaian cultural norms of communal trust, religious faith, and deference to authority, delivered in local languages with native-level fluency. A defence that relies solely on individual scepticism—“trust no one, verify everything”—is culturally inappropriate in a relational trust society (9). The defence must be as culturally sophisticated as the attack.

Proposed Action: The Ministry of Communications and Digitalisation, in partnership with the Cyber Security Authority, the Bank of Ghana, and the Mobile Network Operators, should launch a sustained, multi-platform national public education campaign under the banner “Verify Before You Trust” with the following design principles:

- **Culturally grounded messaging:** Campaign content would be developed in consultation with Ghanaian creative professionals, linguists, and community leaders, and delivered in all major Ghanaian languages. The messaging would leverage local cultural values—protection of family and community, integrity in financial dealings, the responsibility of the knowledgeable to protect the vulnerable—rather than importing individualistic Western awareness frameworks that are culturally dissonant and consequently ineffective (19).
- **Multi-platform delivery:** The campaign would utilise television, radio, social media (particularly WhatsApp, given its centrality to Ghanaian social infrastructure), SMS broadcasts via MNO platforms, and physical materials distributed through the MoMo agent network and community institutions including churches, mosques, and traditional councils.
- **Single, consistent message:** The campaign would deliver one simple, memorable message—“Verify Before You Trust”—repeated consistently across all platforms and languages over a sustained period of years, not weeks. MTN Ghana, Telecel Ghana, and AT Ghana would be required to include the campaign message in all customer-facing communications.
- **Integration with the NM3F:** The campaign would be operationally linked to the NM3F proposed in Recommendation 1, ensuring that campaign messaging is updated in real time to reflect emerging fraud patterns identified through the NM3F’s threat intelligence platform.
- **Measurable outcomes:** The campaign would establish baseline and longitudinal metrics—including public awareness surveys, smishing click-through rates from simulated campaigns, and fraud report volumes—to enable evidence-based evaluation of effectiveness and iterative refinement of messaging.

Evidence Base: The culture-centered prevention model of Adewopo et al. (19) and the cross-cultural trust theory of Alhasan (9) provide the theoretical foundation. The MTN Ghana awareness campaign results reported by Adongo Mendel (2), while requiring verification against primary sources, suggest that MNO-led awareness initiatives can achieve meaningful improvements in fraud recognition. The campaign would operationalise these findings at national scale.

6.6 Prioritisation and Implementation Roadmap

The five recommendations are presented in order of priority, reflecting both their individual impact and their interdependencies. Recommendation 1 (NM3F) is foundational: without effective inter-agency coordination, Recommendations 2 through 5 cannot be implemented effectively. Recommendation 4 (liability architecture) is the most urgent from a consumer protection perspective and should be implemented in parallel with Recommendation 1. Recommendations 3 (digital lending directive) and 5 (public education campaign) are medium-term interventions that depend on the coordination infrastructure established by Recommendation 1.

Implementation should be phased as follows:

- **Phase 1 (0–6 months):** Establish the NM3F. Issue the joint BoG–NCA liability directive. Begin consultation on the digital lending data ethics directive.
- **Phase 2 (6–12 months):** Issue the digital lending data ethics directive. Amend the BoG Cyber and Information Security Directive to mandate psychological inoculation training. Launch the “Verify Before You Trust” campaign pilot in two regions.
- **Phase 3 (12–24 months):** Full national rollout of the public education campaign. First-year review of NM3F operational effectiveness. First compliance audit of the digital lending data ethics directive.

6.7 Chapter Summary

The five recommendations presented in this chapter form an integrated, evidence-based response to the human-centric threat landscape documented throughout this report. They are not generic cybersecurity prescriptions but specific, sourced, and actionable proposals grounded in the Ghanaian context. The National Mobile Financial Fraud Coordination Framework addresses the regulatory fragmentation that enables threats to flourish. The psychological inoculation training mandate targets the human vulnerability that social engineering exploits. The digital lending data ethics directive closes the regulatory gap that allows data weaponization to continue. The liability architecture fixes the injustice that erodes trust in the digital financial system. And the national public education campaign builds the long-term cultural resilience that no technical control can provide.

Together, these recommendations constitute a shift from a reactive, fragmented, technology-centric security posture to a proactive, coordinated, human-centric defence. The final chapter concludes this report with a synthesis of findings and a forward-looking assessment of the evolving threat landscape.

Chapter 7

Conclusion

This report set out to answer a single question: what is the greatest cyber threat to Ghana’s financial services sector? The answer, established across six chapters of evidence, is unequivocal. The greatest threat is not a piece of code. It is not a zero-day exploit, a ransomware variant, or a sophisticated Advanced Persistent Threat actor probing the perimeter of a core banking system. The greatest threat is the systematic, industrialised manipulation of human beings—enabled by vulnerable mobile platforms, exploited by predatory digital lending models, and allowed to fester in a fragmented regulatory landscape.

The four pillars of this report—the Mobile Money ecosystem, social engineering, digital lending, and the regulatory framework—are not separate problems requiring separate solutions. They are a single, interconnected ecosystem of risk. The SIM card that serves as the root of trust for a MoMo wallet is the same SIM card that a social engineer hijacks through a vishing call to a telecom agent. The digital lending platform that extends credit to a financially vulnerable borrower is the same platform that harvests that borrower’s contact list for debt shaming. The regulatory body that licenses the bank is not the body that regulates the telecom provider, and neither is the body that enforces data protection—and the fraudster exploits the seams between them all.

The evidence assembled in this report demonstrates that this ecosystem is not static. Generative artificial intelligence is accelerating every dimension of the threat. AI-enabled elements were present in nearly half of all cybercrime cases in Ghana by the first half of 2025, and AI-enabled attacks cause nearly three times the financial damage of conventional attacks (3). Deepfake voice cloning is exploiting the voice-based trust that underpins Ghanaian agent and peer-to-peer transactions. Synthetic identities generated by adversarial networks are defeating the KYC verification systems that were supposed to be the gatekeepers of the formal financial system. The threat is evolving faster than the regulatory and defensive apparatus designed to contain it.

Yet this report is not a counsel of despair. The five strategic recommendations presented in Chapter 6—a National Mobile Financial Fraud Coordination Framework, mandated psychological inoculation training, a binding Digital Lending Data Ethics Directive, a fixed liability architecture, and a culturally grounded national public education cam-

paign—constitute a coherent, evidence-based, and implementable response. They are not generic prescriptions imported from international best-practice frameworks. They are specific to Ghana’s institutional architecture, cultural context, and threat profile. They address the structural vulnerabilities identified in Chapter 2, the psychological attack vectors dissected in Chapter 3, the predatory data practices exposed in Chapter 4, and the regulatory fragmentation critiqued in Chapter 5.

The central insight of this report—that the human being, not the computer, is the primary attack surface—carries a corollary. If the vulnerability is human, the defence must be human too. Technical controls are necessary but insufficient. A firewall cannot stop a grandmother in Kumasi from sharing her MoMo PIN with a fraudster who speaks fluent Twi and claims to be from MTN customer service. Only a defence architecture that places human psychology, cultural context, and institutional accountability at its centre can hope to match the sophistication of the threat.

The digital inclusion paradox that Arthur et al. (3) identify—the tragic irony that the infrastructure of financial inclusion simultaneously enables financial predation—is not an argument against digital finance. It is an argument for digital finance done differently. Ghana’s Mobile Money revolution has brought millions into the formal financial system. The challenge now is not to retreat from that achievement but to secure it. To build a regulatory architecture that closes the liability vacuum. To mandate training that inoculates users against the specific psychological tactics deployed against them. To regulate digital lending not as a separate sector but as an integral part of the mobile financial ecosystem, with data ethics obligations that match its data harvesting capabilities. And to launch a public education campaign that speaks to Ghanaians in their own languages, drawing on their own cultural values, about the threats that target their own communities.

The threat landscape described in this report will not remain static. By the time this document is read, new AI capabilities will have emerged, new fraud typologies will have been documented, and new regulatory responses will have been proposed. The specific recommendations in Chapter 6 will require adaptation as the threat evolves. But the core principle—that Ghana’s cybersecurity posture must be reoriented from a technology-centric to a human-centric paradigm—is enduring.

The battlefield is the mobile phone. The weapon is the manipulated human mind. The frontier is the predatory lending app. The shield is the fragmented regulatory framework. The solution is a coordinated, culturally grounded, psychologically informed, human-centric defence.

The work of building that defence begins now.

Bibliography

- [1] Mogaji, E., & Nguyen, N. P. (2022). The dark side of mobile money: Perspectives from an emerging economy. *Technological Forecasting & Social Change*, 185, 122045.
- [2] Adongo Mendel. (n.d.). *Mobile money social engineering attacks in African countries: A survey* [Unpublished survey/working paper]. University of Mines and Technology.
- [3] Arthur, J. S., Akonor, E. T., & Obeng, A. S. (2025). The digital inclusion paradox: Mobile money expansion, generative AI, and the escalating architecture of cybercrime in Ghana [Preprint]. *SSRN*. <https://ssrn.com/abstract=6556708>
- [4] Kim, M., Suh, J., & Kwon, H. (2022). A study of the emerging trends in SIM swapping crime and effective countermeasures. In *2022 IEEE/ACIS 7th International Conference on Big Data, Cloud Computing, and Data Science (BCD)* (pp. 240–245). IEEE.
- [5] Anunobi, S., Memon, M., Howard, C., & Lawal, B. (2024). Mitigating security vulnerabilities in offline USSD payments in non-smartphones. In *2024 11th International Conference on Software Defined Systems (SDS)* (pp. 42–49). IEEE.
- [6] Lamoyero, Z., & Fajana, O. (2023). Exposed: Critical vulnerabilities in USSD banking authentication protocols. In *2023 IEEE International Conference on Cyber Security and Resilience (CSR)* (pp. 275–280). IEEE.
- [7] Mutabazi, P., Ndalama, F. E., Taenaka, Y., & Kadobayashi, Y. (2025). Mitigating SIM swap fraud in mobile money platforms in SSA: A zero trust identity and access management architecture. In *2025 8th ICOIACT* (pp. 220–225). IEEE.
- [8] Zaoui, M., Belfaik, Y., Sadqi, Y., Maleh, Y., & Ouazzane, K. (2024). A comprehensive taxonomy of social engineering attacks and defense mechanisms. *IEEE Access*, 12, 72224–72241.
- [9] Alhasan, I. (2023). *Human factors in cybersecurity: A cross-cultural study on trust* [Doctoral dissertation, Purdue University]. Purdue University Graduate School.
- [10] Karadsheh, L., Alryalat, H., Alqatawna, J., Alhawari, S. F., & Al Jarrah, M. A. (2022). The impact of social engineer attack phases on improved security countermeasures. *International Journal of Digital Crime and Forensics*, 14(1), 1–26.
- [11] Chhabra Roy, N., & Prabhakaran, S. (2023). Sustainable response system building against insider-led cyber frauds in banking sector: A machine learning approach. *Journal of Financial Crime*, 30(1), 48–85.
- [12] Wrigley, J. (2018). *Exploring the causes and defenses of social engineering in developing nations: Using Ghana as a case study* (Publication No. 10828888) [Doctoral dissertation, Colorado Technical University]. ProQuest.

- [13] Spangler, M. J. (2021). *Business email compromise: Impacts and strategies for protecting against social engineering attacks* [Master's capstone project, Utica College].
- [14] Nartey, C., Quaye, S., & Dawson, M. (2025). A case study on Ghana's cybersecurity posture: Strengthening cybercrime enforcement and regional cooperation in West Africa. *RAIS Conference Proceedings*, August 7–8, 2025.
- [15] Sambo, M. Q. A., Yusuf, A., AbdulHamid, S., Nwokoro, I., Aliyu, A., & Sadiq, A. U. (2026). AI-driven identity theft and fraud detection in FinTech systems: A systematic literature review (2016–2025). *2026 International Conference on AI Innovations and Industry (ICAIII)*. IEEE.
- [16] Upadhyaya, R., Weitzberg, K., & Bonyo, L. (2025). Digital credit providers, regulatory frameworks, and structural power: A case study of digital microcredit regulation in Kenya. *Finance and Society*, 11, 186–208.
- [17] Opuni-Frimpong, J., Dzorka, M. A., & Boadi, I. (2025). Governance's role in bank performance: cybersecurity committee assessment. *Journal of Financial Reporting and Accounting*, 23(2), 788–810.
- [18] Mensah, G. B., Mijwil, M. M., & Abotaleb, M. (2025). Assessing Ghana's Cybersecurity Act 2020: AI training and medical negligence cases. *Journal of Integrated Engineering and Applied Sciences*, 3(1), 175–182.
- [19] Adewopo, V. A., Azumah, S. W., Yakubu, M. A., Gyamfi, E. K., Ozer, M., & Elsayed, N. (2025). Comprehensive analytical review of cybercrime and cyber policy in West Africa. *Journal of Electrical Systems and Information Technology*, 12, 20.
- [20] Bello, A., Uzoma, C., Atadoga, S., Dehumo, A. P., & Akintola, A. S. (2026). Responsible AI in financial identity verification and risk mitigation. *Discover Sustainability*.
- [21] Anane-Donkor, L., & Dei, D.-G. J. (2026). Adoption and implementation of emerging digital financial solutions for financial inclusion in Ghana. *Discover Global Society*, 4, 136.
- [22] Egbobamwonyi-Bedaux, J. U. (2026). Reimagining sovereignty in the Global South: digital dependency, AI and cybersecurity. *Third World Quarterly*. <https://doi.org/10.1080/01436597.2026.2690081>
- [23] Carter, W. A. (2017). *Forces shaping the cyber threat landscape for financial institutions* (SWIFT Institute Working Paper No. 2016-004). SWIFT Institute / CSIS.
- [24] Zibo, B. S., & Nweke, O. C. (2024). The intersection of political power and data exploitation: Implications for privacy and security in Ghana's digital age. *Journal of Social and Political Sciences*, 7(4), 89–96.